SECS/GEM Reference Manual

for the

Cooling Equipment System

Manual Information

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1.0.0	1st September 2023	Muhammad Adib	Initial release.

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1 INTRODUCTION

1.1 Application

This manual has been prepared to define the communication functions available between PENTAMASTER TECHNOLOGY SDN. BHD., the manufacturer of semiconductor manufacturing equipment Cooling Equipment Machine, which is hereafter called as "the equipment" and its host computer.

Unless otherwise specified, the descriptions made in this manual conform in general to the descriptions made in the SEMI E4-91, SEMI E5-93, and SEMI E30-94.

1.2 SECS Standard

The SEMI Equipment Communications Standard (SECS) is published by Semiconductor Equipment and Materials International (SEMI). It defines a computer to computer communications interface between a unit of factory Equipment and a Host Computer.

This Equipment complies to either SECS-I (E4, serial communication) or HSMS (E37, network communication), and to the generic portions of the SECS-II (E5) standard, and implements an appropriate subset of the SECS-II standard messages required by GEM (E30).

Cooling Equipment Machine complies to the following versions of standards:

Item	Standard
SECS-I	**SEMI E4-91** **SEMI Equipment Communications Standard 1, Message Transfer**
SECS-II	**SEMI E5-93** **Equipment Communications Standard 2, Message Contents**
GEM	**SEMI E30-94** **Generic Model for Communications And Control of SEMI Equipment**
HSMS	**SEMI E37-95** **High-Speed SECS Message Services (HSMS), Generic Services**
HSMS-SS	**SEMI E37.1-95** **High-Speed SECS Message, Single-Session Mode (HSMS-SS)**

1.3 GEM Compliance Statement

This GEM Compliance Statement accurately indicates for each capability whether it has been implemented, and whether the implementation is in a GEM compliant manner. The GEM Compliance Statement for the Equipment is given in the Table as below.

Table: GEM Compliance Statement

Fundamental GEM Requirements	Implemented	GEM Compliant
State Models	Yes	Yes
Equipment Processing States	Yes	
Host Initiated S1F13/S1F14 Scenario	Yes	
Event Notification	Yes	
On-Line Identification	Yes	
Error Messages	Yes	
Documentation	Yes	
Control (Operator Initiated)	Yes	
Additional Capabilities	**Implemented**	**GEM Compliant**
Establish Communications	Yes	Yes
Dynamic Event Report Configuration	Yes	Yes
Variable Data Collection	Yes	Yes
Trace Data Collection	Yes	Yes
Status Data Collection	Yes	Yes
Alarm Management	Yes	Yes
Remote Control	Yes	Yes
Equipment Constants	Yes	Yes
Process Program Management	Yes	Yes
Material Movement	Yes	Yes
Equipment Terminal Services	Yes	Yes
Clock	Yes	Yes
Limits Monitoring	Yes	Yes
Spooling	Yes	Yes
Control (Host Initiated)	Yes	Yes

1.4 SECS-I Connection

In case of serial (RS-232) communication, the communication between the equipment and the host conforms to the SECS-I/II as defined by the SEMI (Semiconductor Equipment and Materials International).

1.4.1 SECS-I Physical Connection

The following set values are used as circuit parameters:

Item	Description
Electrical I/F	**EIA RS-232-C complied**
Connector	**<ISO 2110-1980(D-sub) 25pin [FEMALE]>**
Connector location	**<Connector location>**
Signal pins	**Pin# Description** **1 shield** **2 data transmit (output)** **3 data receive (input)** **7 signal ground**
Data rate	**9600 bps (changeable) [300-19200]**
Characters	**Start bit = 1bit** **Data bit = 8bit (LSB-MSB)** **Stop bit = 1bit** **Parity bit = none**

1.4.2 SECS-I Parameters

The following set values are used as circuit control parameters:

Parameters	Settable range	Default	Description
Device ID	**0 - 7FFF (hex)**	**0**	**Identifier assigned to the equipment.**
T1	**0.1 - 10.0 (sec)**	**0.5 (sec)**	**Detects an interruption between characters.**
T2	**0.2 - 25.0 (sec)**	**10 (sec)**	**Detects a lack of protocol response.**
T3	**1 - 120 (sec)**	**45 (sec)**	**Detects a lack of reply message.**
T4	**1 - 120 (sec)**	**45 (sec)**	**Detects an interruption in a multi-block message.**
RTY	**0 - 31 (times)**	**3 (times)**	**The maximum of send retries allowed.**

1.5 HSMS Connection

In case of network (EtherNet) communication, the communication between the equipment and the host conforms to the HSMS, HSMS-SS and SECS-II as defined by the SEMI (Semiconductor Equipment and Materials International).

1.5.1 HSMS Physical Connection

If HSMS communication port is equipped, the equipment follows the following connection type:

Item	Description
Electrical I/F	**IEEE 802.3 complied**
Connector	**100Base**
Connector location	**Panel PC RJ45**

1.5.2 HSMS Parameters

The following set values are used as circuit control parameters:

Parameters	Settable range	Default	Description
Connect Mode	**PASSIVE/ACTIVE**	**PASSIVE**	**HSMS connection mode.**
IP Address	**Determined by TCP/IP conventions**	**N/A**	**Remote IP address when the equipment is ACTIVE; Local IP address when the equipment is PASSIVE.**
Port Number	**Determined by TCP/IP conventions**	**N/A**	**The Port number when the equipment is ACTIVE/PASSIVE.**
T3	**1 - 120 (sec)**	**45 (sec)**	**Detects a lack of reply message.**
T5	**1 - 240 (sec)**	**60 (sec)**	**Connect Separation Timeout.**
T6	**1 - 240 (sec)**	**5 (sec)**	**Control Transaction Timeout.**
T7	**1 - 240 (sec)**	**10 (sec)**	**NOT SELECTED Timeout.**
T8	**1 - 120 (sec)**	**5 (sec)**	**Network Inter-character Timeout.**

2 STATE MODELS

2.1 Establish Communications State Model

This section describes the model of established communication state the equipment may provide.

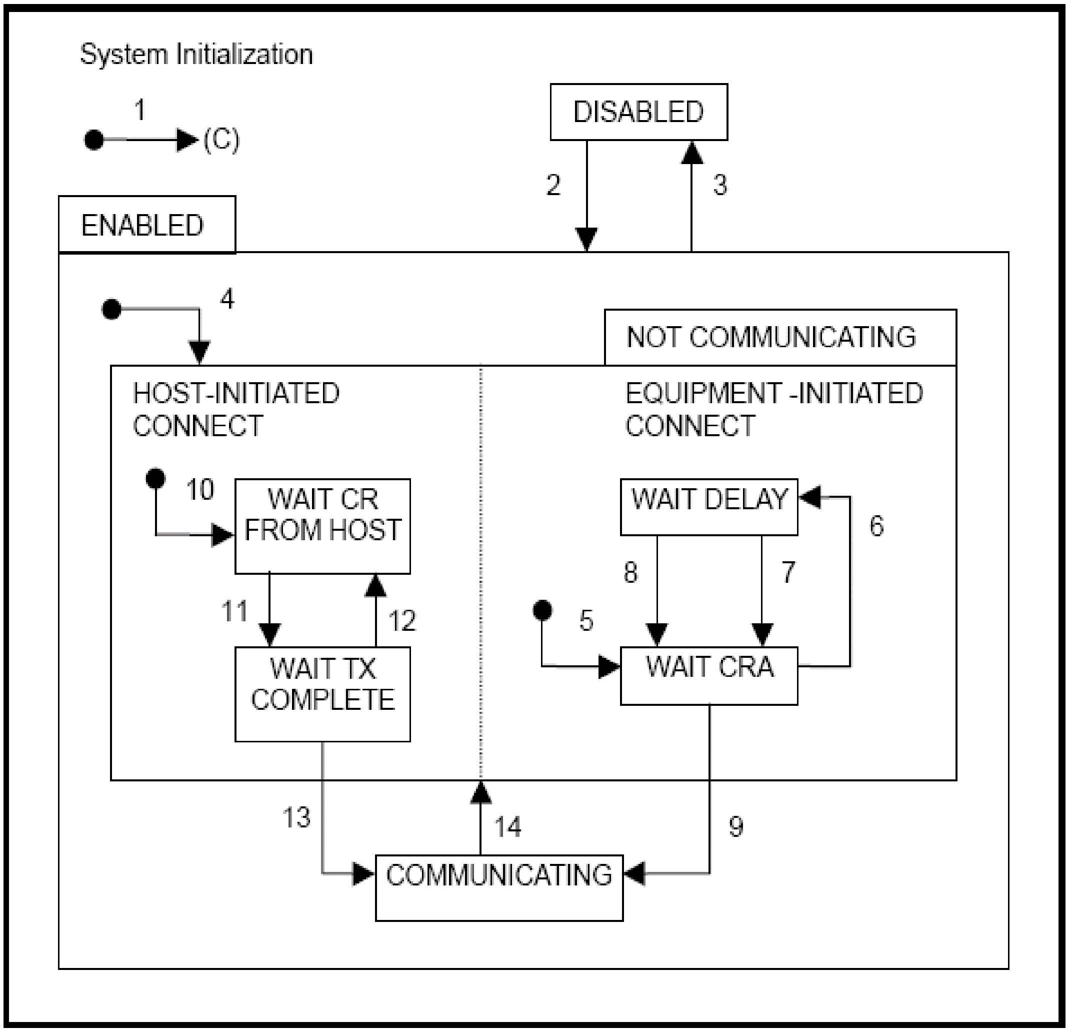
1) Description of Functions

The equipment manages the state of communication line with the host as the model of establish communication state. The established communication state model provides the functions conforming to those specified in the E30.

2) State Definition

STATE NAME	Description
DISABLED	The state where communication with the host is disabled. The messages from the host may be received but all of them will be ignored.
ENABLED	The state where communication with the host is enabled.
NOT COMMUNICATING	Waits for establishment of communication (S1F13/F14). Waiting the S1F13 from the host. Message from the host other than S1F13 will be discarded when received. A message created by the equipment in this state, such as event report, will be discarded unless it is not spool active. For details, see the spooling description.
EQUIPMENT – INITIATED CONNECT	Waits for an Establish Communications message from the host, and activates upon HOST-INITIATED CONNECT.
WAIT CRA	Sends S1F13 to the host and waits for S1F14.
WAIT DELAY	Monitors the transmit interval timer of S1F13.
HOST-INITIATED CONNECT	Waits for an Establish Communications message from the host, and activates upon EQUIPMENT-INITIATED CONNECT.
WAIT CR FROM HOST	Waits for S1F13 from the host.
WAIT TX COMPLETE	Waits for completion of S1F14 in response to the S1F13 received from the host.
COMMUNICATING	Communication with the host has been established and enabled.

3) State Chart



4) Transition Table

#	Current State	Trigger	New State	Action	Comment
1	-	System Initialize	Default	None	New state will be specified by equipment parameter.
2	DISABLED	Operator switches from DISABLED to ENABLED.	ENABLED	None	SECS-II communications are enabled.
3	ENABLED	Operator switches from ENABLED to DISABLED.	DISABLED	None	SECS-II communications are prohibited.
4	(Entry to ENABLED)	Any entry to ENABLED state.	NOT COMMUNICATI NG	None	May enter from system initializa-tion or through operator switch.
5	(Entry to EQUIPMENT INITIATED CONNECT)	Any entry to NOT COMMUNICATIN G.	WAIT CRA	Set CommDelay timer "expired". Send S1F13.	Initialize communications.
6	WAIT CRA	Connection transaction failure.	WAIT DELAY	Initialize CommDelay timer.	Wait for timer to expire.
7	WAIT CRA	CommDelay timer expired.	WAIT CRA	Send S1F13.	May receive S1F13 from host.
8	WAIT DELAY	Received a message other than S1F13 from host.	WAIT CRA	Discard the message.	No reply.
9	WAIT CRA	Received expected S1F14 with COMMACK=0.	COMMUNICATI NG	None	Communications are established.
10	(Entry to HOST INITIATED CONNECT	Any transition to NOT COMMUNICATIO NG.	WAIT CR FROM HOST	None	Wait for S1F13 from host.
11	WAIT CR FROM HOST	Received S1F13 from host.	WAIT TX COMPLETE	Send S1F14 with COMMACK=0.	
12	WAIT TX COMPLETE	S1F14 transmission failed.	WAIT CR FROM HOST	None	Wait for S1F13 from host.
13	WAIT TX COMPLETE	S1F14 transmission completed successfully.	COMMUNICATI NG	None	Communications are established.
14	COMMUNI-CATING	Communication failed.	NOT COMMUNICATI NG	Dequeued messages may be placed in spool buffer as appropriate	If spooling function is provided.

2.2 Control State Model

This section describes the control state the equipment provides.

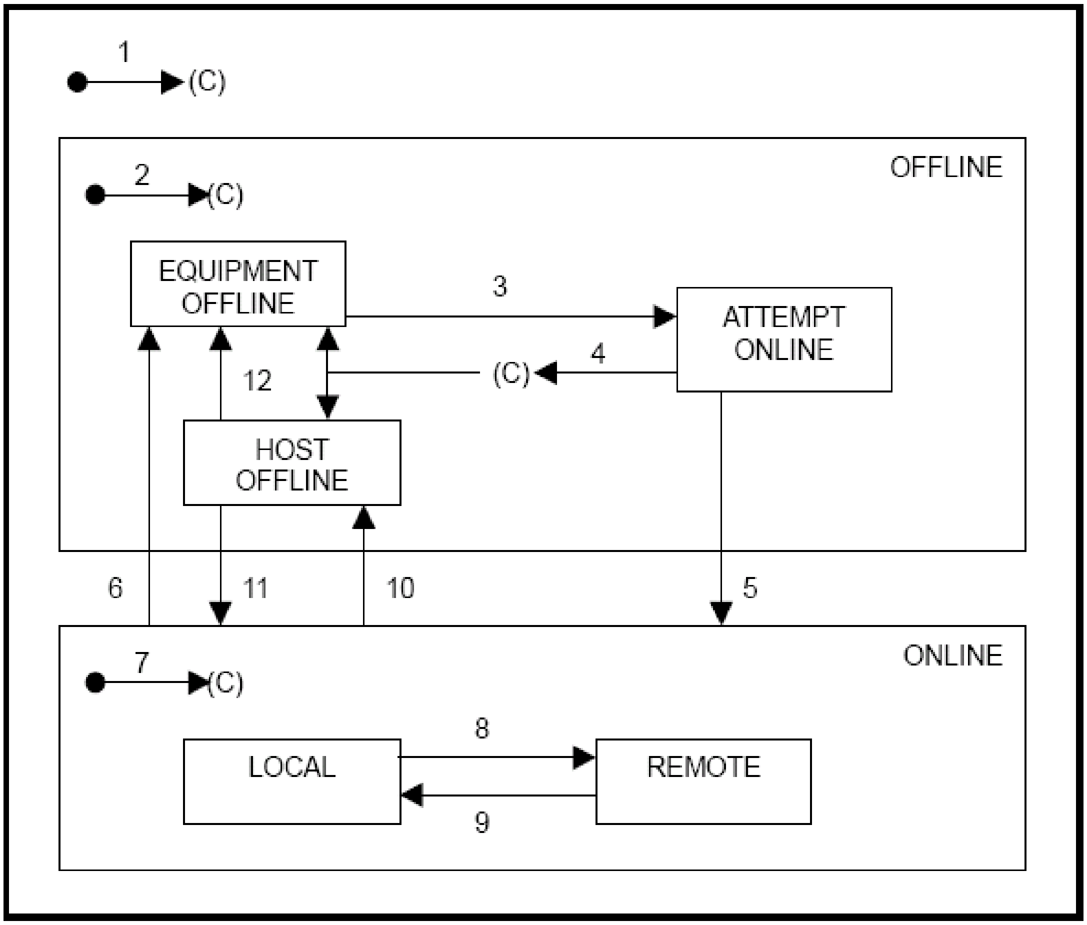
1) Description of Functions

The equipment provides the control state model as operating mode to be manipulated by host or operator.

2) State Definition

STATE NAME		Description
OFFLINE		Communication is enabled. Equipment may be manipulated by Operator only. Respond any message other than S1F13 and F17 by aborting (SxF0).
	EQUIPMENT OFFLINE	Equipment configured to default to OFFLINE. Wait for switching ONLINE/OFFLINE.
	HOST OFFLINE	Host selects to be offline. Wait for S1F17 from host, or operator switching to OFFLINE.
	ATTEMPT ONLINE	Attempting to transit to ONLINE.
ONLINE		Able to operate by exchanging SECS messages.
	LOCAL	Operator is enabled to manipulate Equipment directly. Host communication is limited. For items limited, see Remote Control section.
	REMOTE	Equipment is operated by Host. Operator's manipulation is limited. For items limited, see Remote Control section.

3) State Chart



4) Transition Table

#	Current State	Trigger	New State	Comment (event)
1	-	System Initialize	default	New state depends on Equipment parameters.
2	-	Entry to OFFLINE state.	default	New state depends on Equipment parameters.
3	EQUIP. OFFLINE	Operator selects ONLINE switch.	ATTEMPT ONLINE	S1F1 is sent.
4	ATTEMPT ONLINE	S1F1/F2 transmission failed.	EQUIP. or HOST OFFLINE	New state depends on Equipment parameters.
5	ATTEMPT ONLINE	Equipment receives expected S1F2.	ONLINE	Host is notified of equipment has turned to ONLINE.
6	ONLINE	Operator selects OFFLINE switch.	EQUIP. OFFLINE	Report GemControlStateOFFLINE event.
7	-	Entry to ONLINE state.	default	Defaults upon entry to ONLINE depend on Equipment console switch setting. Report "Control state" event.
8	LOCAL	Operator selects REMOTE switch.	REMOTE	Report GemControlStateREMOTE event.
9	REMOTE	Operator selects LOCAL switch.	LOCAL	Report GEMControlStateLOCAL event.
10	ONLINE	Equipment receives S1F15 from host.	HOST OFFLINE	Report GemControlStateOFFLINE event.
11	HOST OFFLINE	Equipment receives S1F17 from host.	ONLINE	
12	HOST OFFLINE	Operator selects OFFLINE switch.	EQUIP. OFFLINE	Report GemControlStateOFFLINE event.

2.3 Process State Model

This section describes the process state models provided by the equipment.

1) Description of Functions

This state represents current availability of the equipment.

2) State Definition

STATE NAME	Description
RUNNING	Equipment is now running.
STOP	Equipment is now stop.

2.4 Alarm State Model

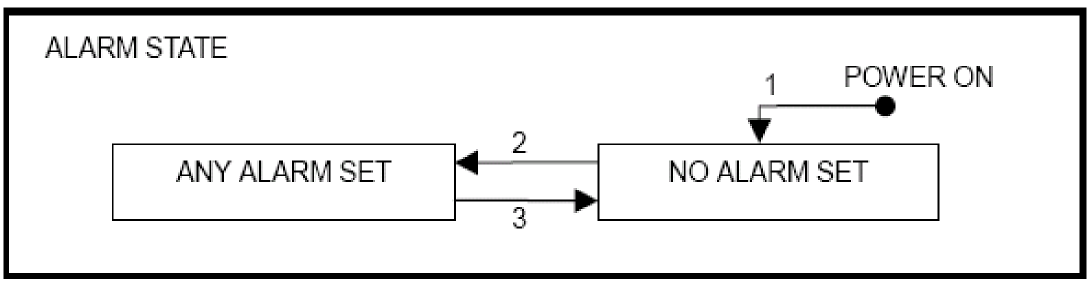
1) Description of Functions

This state represents current alarm condition of the equipment.

2) State Definition

STATE NAME	Description
NO ALARM SET	The equipment has no alarm.
ANY ALARM SET	The equipment has at least one alarm.

3) State Chart



4) Transition Table

#	Current State	Trigger	New State	Comment (event)
1	Undefined	Power ON. There is no alarm.	NO ALARM SET	none.
2	NOT ALARM SET	An alarm occurred.	ANY ALARM SET	
3	ANY ALARM SET	All alarm has been cleared.	NO ALARM SET	

2.5 Limit Monitoring State Model

1) Description of Functions

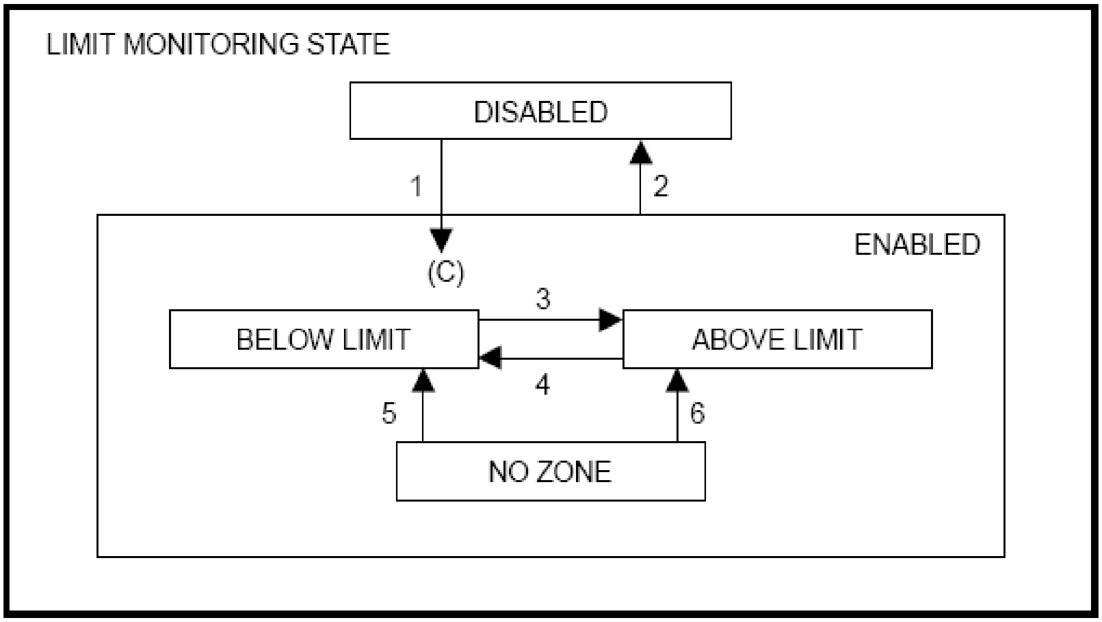
The equipment provides the following as limit monitoring function.

- The maximum number of SVs, which can be monitored in same time is 4.
- Host is capable of specifying 7 limit ID's (LIMITID) for one SV.
- Equipment will constantly monitor the LIMITID specified by the host. If the SV experiences a change exceeding a LIMITID, the equipment sends it as a GemLimitZoneTransition event report.
- A LIMITID has the band width between the UPPERDB and LOWERDB.
- Equipment's SV monitor interval is set to one second, and a change occurring in shorter time span may not be monitored.
- The SV's specifiable of the limit values are defined separately.

2) State Definition

STATE NAME		Description
DISABLED		Limits are not specified.
ENABLED		Limits are specified.
	BELOW LIMIT	Specified variable has decreased to the level equal to LOWERDB or below.
	ABOVE LIMIT	Specified variable has increased to the level equal to UPPERDB or above.
	NO ZONE	Specified variable is in neither UPPER ZONE nor LOWER ZONE. The previously set value of the specified variable is not stable, and remains between UPPERDB and LOWERDB. This phenomenon may occur at the time of equipment startup, or when a new limit is specified.

3) State Chart



4) Transition Table

#	Current State	Trigger	New State	Comment (event)
1	DISABLED	Limit attributes (limit value) defined with S2F45.	ENABLED	Substate of ENABLED is determined by the current value of the monitored variable.
2	ENABLED	Limit attributes set to undefined w/S2F45.	DISABLED	none.
3	BELOW LIMIT	Variable increased to be ≧ UPPERDB.	ABOVE LIMIT	Zone transition and event occurred. TransitionType =0
4	ABOVE LIMIT	Variable decreased to be ≦ LOWERDB.	BELOW LIMIT	Zone transition and event occurred. TransitionType =1
5	NO ZONE	Variable decreased to be ≦ LOWERDB.	BELOW LIMIT	Zone transition and event occurred. TransitionType =1
6	NO ZONE	Variable increased to be ≧ UPPERDB.	ABOVE LIMIT	Zone transition and event occurred. TransitionType =0

2.6 Spooling State Model

1) Description of Functions

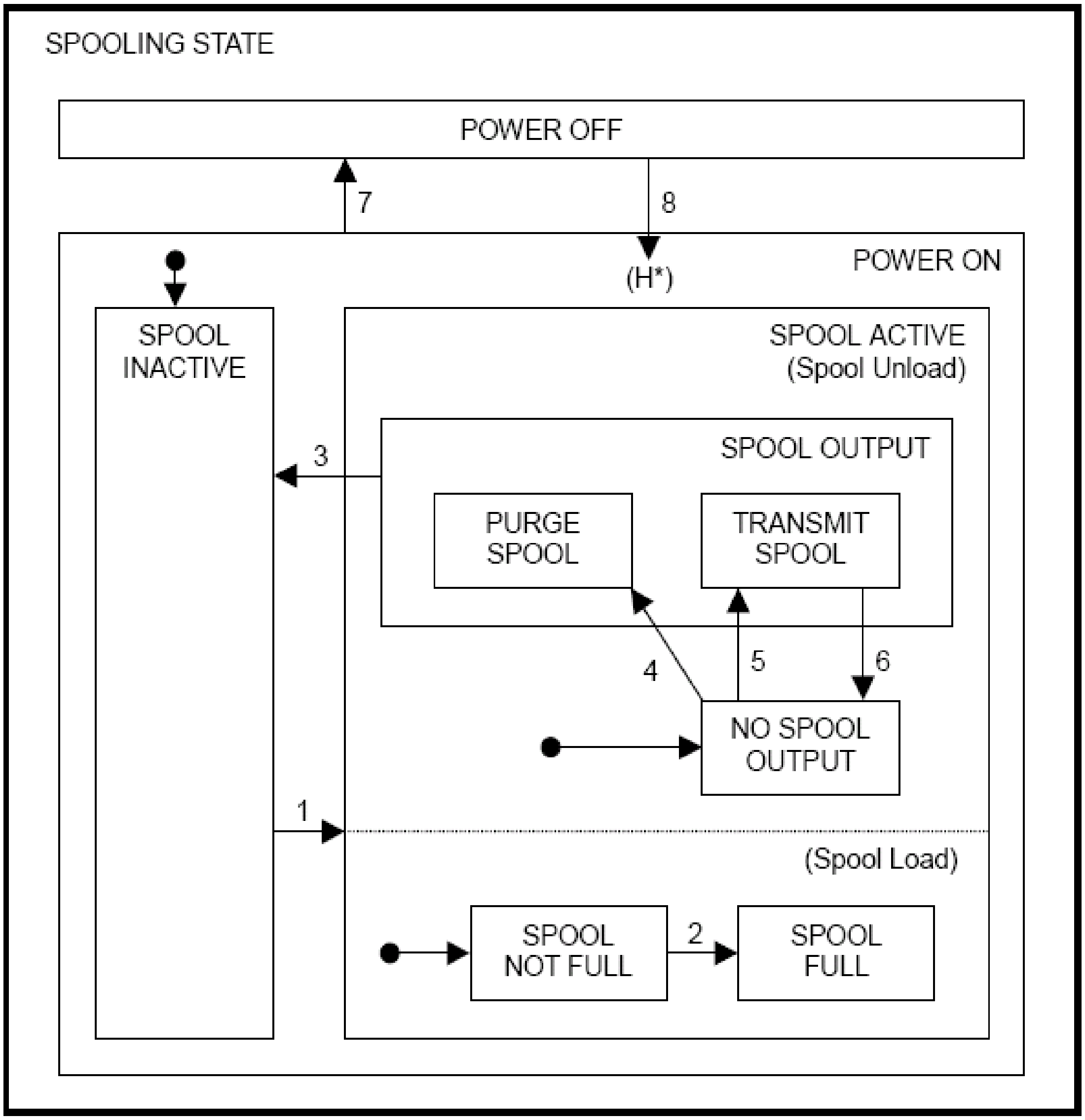
The equipment provides the following functions for queuing messages intended for the host during times of communication failure and subsequently deliver them when communication is restored:

- Once the communication with the host is established, any message failed to send to the host will be stored in the spool area.
- The messages in the spool area will be transmitted to the host when it requests.
- Action to take when the spool area has become full may be specified on the equipment constant "OverWriteSpool".
- The spooled data will be retained in non-volatile storage when the equipment power is turned off.
- The messages to spool are limited to the primary messages issued by the equipment excluding the Stream 1 messages. The host may specify what messages to spool.
- If a block comprising a multi-block message failed to be transmitted, the entire message will be spooled.

2) State Definition

STATE NAME	Description
SPOOL INACTIVE	Normal operating mode. No spooling occurs.
SPOOL ACTIVE	Messages specified in S2F43 are directed to the spool area for loading or unloading.
SPOOL UNLOAD	Moves messages out of the spool.
SPOOL OUTPUT	Removes messages out of the spool to transmit or purge.
PURGE SPOOL	Discards messages in the spool.
TRANSMIT SPOOL	Transmits messages in the spool.
NO SPOOL OUTPUT	Waits for the host to request the message transmission, and no messages are removed from the spool.
SPOOL LOAD	Enters messages into the spool area.
SPOOL NOT FULL	Spooling is enabled.
SPOOL FULL	Overwriting messages in the spool is enabled when set to OverWriteSpool=1. If OverWriteSpool=0, further spooling is disabled unless the empty area is available.

3) State Chart



4) Transition Table

#	Current State	Trigger	New State	Comment (event)
1	SPOOL INACTIVE	Communication failure detected.	SPOOL ACTIVE	SPOOL LOAD/ UNLOAD become active and message that could not be sent will be spooled. SPOOL active event.
2	SPOOL NOT FULL	SPOOL area is full.	SPOOL FULL	Set SpoolFullTime. Alert Operator that spool is full.
3	SPOOL OUTPUT	SPOOL area is empty.	SPOOL INACTIVE	SPOOL deactivated event.
4	NO SPOOL OUTPUT	S6F23 received with RSDC=1.	PURGE SPOOL	Purge the message in SPOOL area.
5	NO SPOOL OUTPUT	S6F23 received with RSDC=0.	TRANSMIT SPOOL	Transmit messages in SPOOL area.
6	TRANSMIT SPOOL	Communication failure or MaxSpoolTransmit reached.	NO SPOOL OUTPUT	If communication failure, the event Spool Transmit Failure has occurred.
7	POWER ON	Power off.	POWER OFF	Spooling context maintained in nonvolatile storage.
8	POWER OFF	Power on.	POWER ON	Spooling context restored from non-volatile memory storage.

3 SECS MESSAGE SUMMARY

Cooling Equipment system supports the following SECS II messages.

3.1 Host To Equipment

Table 1: Supported SECS Messages, Host->Equipment

Host Primary	Equipment Reply	Description
(Any)	SxF0	Off-line Abort Reply
S1F1	S1F2	Are You There Request
S1F3	S1F4	Selected Equipment Status Request
S1F11	S1F12	Status Variable Namelist Request
S1F13	S1F14	Establish Communication Request
S1F15	S1F16	Request Off-line
S1F17	S1F18	Request On-line
S2F13	S2F14	Equipment Constant Request
S2F15	S2F16	New Equipment Constant Send
S2F17	S2F18	Date and Time Request
S2F23	S2F24	Trace Initialize Send
S2F25	S2F26	Loopback Diagnostic Request
S2F29	S2F30	Equipment Constant Namelist Request
S2F31	S2F32	Date and Time Send
S2F33	S2F34	Define Report
S2F35	S2F36	Link Event Report
S2F37	S2F38	Enable/Disable Event Report
S2F39	S2F40	Multi-Block Inquire
S2F41	S2F42	Host Command Send
S2F43	S2F44	Reset Spooling Streams and Function
S2F45	S2F46	Define Variable Limit Attributes
S2F47	S2F48	Variable Limit Attributes Request
S5F3	S5F4	Enable/Disable Alarm Send
S5F5	S5F6	List Alarms Request
S5F7	S5F8	List Enabled Alarm Request
S6F15	S6F16	Event Report Request
S6F19	S6F20	Individual Report Request
S6F23	S6F24	Request Spooled Data
S7F1	S7F2	Process Program Load Inquire/Grant
S7F3	S7F4	Process Program Send
S7F5	S7F6	Process Program Request
S7F17	S7F18	Delete Process Program Send
S7F19	S7F20	Current EPPD Request
S10F3	S10F4	Terminal Display, Single
S10F5	S10F6	Terminal Display, Multiblock

3.2 Equipment to Host

Table 2: Supported SECS Messages, Equipment→Host

Equipment Primary	Host Reply	Description
(Any)	SxF0	Off-line Abort Reply
S1F1	S1F2	Are You There Request
S1F13	S1F14	Establish Communications Request
S2F17	S2F18	Date and Time Request
S5F1	S5F2	Alarm Report Send
S6F1	S6F2	Trace Data Send
S6F5	S6F6	Multi-block Data Send Inquire
S6F11	S6F12	Event Report Send
S9F1	--	Unrecognized Device ID
S9F3	--	Unrecognized Stream Type
S9F5	--	Unrecognized Function Type
S9F7	--	Illegal Data
S9F9	--	Transaction Timer Time-out
S9F13	--	Conversation Timeout
S9F11	--	Data Too Long
S10F1	S10F2	Terminal Request

4 SECS MESSAGES DETAIL

This section contains the SECS messages referenced in this document. All standard messages are identical to SECS-II and GEM.

Abbreviations: S = **S**ingle block / M = **M**ulti block
H = **H**ost / E = **E**quipment (**QGEMService**)
->valid direction
reply (required)
[reply] (optional)

4.1 Stream 1: Equipment Status

This message stream provides a means of exchanging information about the status of the equipment, including its current mode.

S1F0 Abort Transaction (S1F0) S, H<—>E
Description: Used in lieu of an expected reply to abort a transaction. Function 0 is defined in every stream and has the same meaning in every stream.
Structure: Header only.

S1F1 Are You There Request (R) S, H<->E, reply
Description: Establishes whether the equipment or the host is on line. A function 0 response to this message means that communications are inoperative. In the equipment, a function 0 is equivalent to a timeout on the receive timer after issuing S1F1 to the host.
Structure: Header only.

S1F2 On Line Data (D) S, H<->E
Description: Data signifying that the equipment is alive.
Structure: L,2
<MDLN>
<SOFTREV>
Exception: The host sends a zero-length list to the equipment.

S1F3 Selected Equipment Status Request (SSR) S, H->E, reply
Description: A request to the equipment to report selected status values.
Structure: L,n
<SVID1>
…
<SVIDn>
Exception: A zero-length list means report all SVIDs.

S1F4 Selected Equipment Status Data (SSD) M, H<-E
Description: The equipment reports the value of each SVID requested, in the order requested. The host remembers the names of values requested.
Structure: L,n
<SV1>
…

<SVn>
Exception: If n=0, no response can be made. A zero length list returned for Svi means that SVIDi does not exist.

S1F11 Status Variable Namelist Request (SVNR) S, H->E, reply
Description: A request to the equipment to identify certain status variables.
Structure: L,n
<SVID1>
…
<SVIDn>
Exception: A zero length list means report all SVIDs.

S1F12 Status Variable Namelist Reply (VNRR) M, H<-E
Description: The equipment reports the name and units of the requested status variables to the host.
Structure: L,n
L,3
<SVID1>
<SVNAME1>
<UNITS1>
…
L,3
<SVIDn>
<SVNAMEn>
<UNITSn>

S1F13 Establish Communication Request (CR) S, H<->E, reply
Description: The purpose of this message is to formally establish communications between equipment and host. This is the first message sent following any period of inability to communicate.
Structure: L,2
<MDLN>
<SOFTREV>
Exception: The host sends a zero-length list to the equipment.

S1F14 Establish Communication Request Acknowledge (CRA) S, H<->E
Description: The purpose of this message is to accept or deny the Establish Communication Request (S1F13). MDLN and SOFTREV are on-line data and are valid only if COMMACK = Accepted.
Structure: L,2
<COMMACK>
L,2
<MDLN>
<SOFTREV>
Exception: The host sends a zero-length list for item 2 to the equipment.

S1F15 Request OFF-LINE (ROFL) S, H->E, reply
Description: The host requests an equipment transition to the OFF-LINE state.
Structure: Header only.

S1F16 OFF-LINE Acknowledge (OFLA) S, H<-E
Description: Acknowledge or error.
Structure: <OFLACK>.

S1F17 Request ON-LINE (RONL) S, H->E, reply
Description: The host requests an equipment transition to the ON-LINE state.
Structure: Header only.

S1F18 ON-LINE Acknowledge (ONLA) S, H<-E
Description: Acknowledge or error.
Structure: <ONLACK>.

4.2 Stream 2: Equipment Control and Diagnostics

All messages, which is deal with host control of the equipment. This includes all remote operations and equipment self-diagnostics, but specifically excludes control operations associated with material transfer.

S2F0 Abort Transaction (S2F0) S, H<—E
Description: Same form as S1F0

S2F13 Equipment Constant Request (ECR) S, H->E, reply
Description: Equipment constants, such as those used for calibration, servo gain, alarm limits, data collection mode, plus other values that are infrequently changed, can be obtained using this message.
Structure: L,n
<ECID1>
...
<ECIDn>
Exception: A zero-length list means report all ECVs according to a predefined order.

S2F14 Equipment Constant Data (ECD) M, H<-E
Description: Data response to S2F13 in the order requested.
Structure: L,n
<ECV1>
...
<ECVn>
Exception: If n=0, then no response exists. A zero length ECVi means that ECIDi does not exist.

S2F15 New Equipment Constant Send (ECS) S, H->E, reply
Description: Change one, or more, equipment constants.
Structure: L,n
L,2
<ECID1>
<ECV1>
L,2
<ECIDn>
<ECVn>

S2F16 New Equipment Constant Acknowledge (ECA) S, H<-E
Description: Acknowledge or error.
Structure: <EAC>

S2F17 Date and Time Request (DTR) S, H<->E, reply
Description: Useful to check that the equipment time base is synchronize with the host time base.
Structure: Header only.

S2F18 Date and Time Data (DTA) S, H<->E
Description: Actual time data.
Structure: <TIME>

S2F23 Trace Initialize Send (TIS) S, H->E, reply
Description: Host requests the equipment to periodically sample and report the status variables in the equipment. If the TRID has the same values as previously entered ones, the previous setting will be terminated and the new trace will be initiated. TOTSMP = 0 means to terminate the trace of the specified TRID.
Structure: L,5

<TRID>
<DSPER>
<TOTSMP>
<REPGSZ>
L,n
<SVID1>
…
<SVIDn>

S2F24 Trace Initialize Acknowledge (TIA) S, H<-E
Description: Equipment acknowledges trace initialize data.
Structure: <TIAACK>
Note: The number of the TRID's that may be traced is up to 4. If the trace for more than 4 is specified, it returns 'TIAACK = 2' meaning further number of tracing cannot be performed. The minimum data sampling period rate is 5 second. The maximum number of report group size allows is 4. The maximum number of the SVID allows in a report group size is 100.

S2F25 Loopback Diagnostic Request (LDR) S, H->E, reply
Description: A diagnostic message to checkout protocol and communication circuits.
Structure: <ABS>

S2F26 Loopback Diagnostic Data (LDD) S, H<-E
Description: The echoed binary string.
Structure: <ABS>

S2F29 Equipment Constant Namelist Request (ECNR) S, H->E, reply
Description: This function allows the host to obtain basic information about which equipment constants are available on the equipment.
Structure: L,n

<ECID1>
…
<ECIDn>

Exception: A zero length list means, send information on all ECIDs.

S2F30 Equipment Constant Namelist (ECN) S, H<-E
Description: Return response to S2F29.
Structure: L,n (number of equipment constants)

L,6
<ECID1>
<ECNAME1>
<ECMIN1>
<ECMAX1>
<ECDEF1>

<UNITS1>
L,6
…
…
L,6
<ECIDn>
<ECNAMEn>
<ECMINn>
<ECMAXn>
<ECDEFn>
<UNITSn>

S2F31 Date and Time Set Request (DTS) S, H->E, reply
Description: The host requires the equipment to synchronize its internal clock with the host time based TIME which is given.
Structure: <TIME>

S2F32 Date and Time Set Acknowledge (DTA) S, H<-E
Description: Acknowledge the receipt of time and date.
Structure: <TIACK>

S2F33 Define Report (DR) M, H->E, reply
Description: The purpose of this message is for the host to define a group of reports for the equipment.
Structure: L,2

```
<DATAID>
L,a
    L,2
       <RPTID1>
       L,b
       <VID1>
          ...
       <VIDb>
    ...
    L,2
       <RPTIDa>
       L,c
       <VID1>
          ...
       <VIDc>
```

Exception:

- A zero length list following DATAID deletes all report definitions and associated links.
- A zero length list following RPTID deletes report type RPTID. All CEID links to this RPTID are also deleted.
- Default RPTIDs are reserved for equipment system. New RPTID required for new report definitions.
- Default RPTIDs cannot not delete from the equipment system.

S2F34 Define Report Acknowledge (DRA) S, H<-E
Description: Acknowledge or error.
Structure: <DRACK>

S2F35 Link Event Report (LER) M, H->E, reply
Description: The purpose of this message is for the host to link n reports to an event (CEID). These linked event reports will default to 'disabled' upon linking. That is, the occurrence of an event would not cause the report to be sent until reporting for the event is enabled using S2F37.

```
Structure: L,2
                <DATAID>
                L,a
                     L,2
                        <CEID1>
                        L,b
                        <RPTID1>
                           ...
                        <RPTIDb>
                     ...
                     L,2
                        <CEIDa>
        L,c
                        <RPTID1>
                           ...
                        <RPTIDc>
```

Exception: A zero length list following CEID deletes all report links to that event.

S2F36 Link Event Report Acknowledge (LERA) S, H<-E
Description: Acknowledge or error.
Structure: <LRACK>

S2F37 Enable/Disable Event Report (EDER) S, H->E, reply
Description: The purpose of this message is for the host to enable or disable reporting for a group of events (CEIDs).

```
Structure: L,2
                <CEED>
                L,n #CEIDs
                <CEID1>
                    …
                <CEIDn>
```

Exception: A list of zero length following <CEED> means all CEIDs.

S2F38 Enable/Disable Event Report Acknowledge (EERA) S, H<-E
Description: Acknowledge or error.
Structure: <ERACK>

S2F39 Multiblock Inquire (DMBI) S, H->E, reply
Description: If an S2F33 or S2F35 message is more than one block, this transaction must precede the message.
Structure: L,2

<DATAID>
<DATALENGTH>

S2F40 Multiblock Grant (MBG) S, H<-E
Description: Acknowledge or error.
Structure: <GRANT>

S2F41 Host Command Send (HCS) S, H->E, reply
Description: The host requests the Equipment to perform the specified remote command with the associated parameters.
Structure: L,2

```
    <RCMD> remote command
    L,n n = number of parameters
     L,2
          CPNAME1 name of parameter #1
          CPVAL1 value of parameter #1
       …
       L,2 n = number of parameters
          CPNAMEn name of parameter #n
          CPVALn value of parameter #n
```

S2F42 Host Command Acknowledge (HCA) S, H<-E
Description: Acknowledge host command or error. If a command is not accepted due to one or more invalid parameters (ie, HCACK=3), a list of invalid parameters is returned containing the parameter name and the reason it is invalid.
Structure: L,2

```
    <HCACK>
    L,n n = number of invalid parameters
       L,2
          CPNAME1 name of invalid parameter #1
          CPACK1 reason of invalidity for parameter #1
       ...
       L,2
          CPNAMEn name of invalid parameter #n
          CPACKn reason of invalidity for parameter #n
```

Note:
If there are no invalid parameters, a list of zero length will be sent for item 2.

S2F43 Reset Spooling Streams and Function (RSSF) S, H->E, reply
Description: Host selects specific streams and functions to be spooled whenever spooling is active.
Structure: L,m

```
    L,2
       <STRID1>
       L,n
          <FCNID1>
          …
          <FCNIDn>
    …
    L,2
```

```
<STRIDm>
L,n
    <FCNID1>
    …
    <FCNIDn>
```

Note: If to set off spooling on all messages, set 'm=0'. If to set off spooling on all the messages of a specific stream, set 'n=0'. Spooling cannot be specified on Stream1. The FCNID specified by a STRID will replace the spooling specified by the same STRID previously. Addition of an FCNID cannot be specified.

S2F44 Reset Spooling Acknowledge (RSA) M, H<-E

Description: Equipment acknowledges Host's reset spooling request.

Structure: L,2

```
<RSPACK>
L,m
    L,3
        <STRID1>
        <STRACK1>
        L,n
            <FCNID1>
            …
            <FCNIDn>
    …
    L,3
        <STRIDm>
        <STRACKm>
        L,n
            <FCNID1>
            …
            <FCNIDn>
```

Note: If RSACK=0 and m=0, it indicates no error was found. If n=0, it indicates no functions error for the stream.

S2F45 Define Variable Limit Attributes (DVLA) M, H->E, reply

Description: Host specifies limit monitoring.

Structure: L,2

```
<DATAID>
L,m
    L,2
        <VID1>
        L,n
            L,2
                <LIMITID1>
                L,p
                    <UPPERDB1>
                    <LOWERDB1>
            …
            L,2
                <LIMITIDn>
                L,p
```

```
                        <UPPERDBn>
                        <LOWERDBn>
        …
        L,2
            <VIDm>
            L,n
                L,2
                    <LIMITID1>
                    L,p
                        <UPPERDB1>
                        <LOWERDB1>
                …
                L,2
                    <LIMITIDn>
                    L,p
                        <UPPERDBn>
                        <LOWERDBn>
```

Note: Set 'm=0' to cancel all of the limit monitoring. Set 'n=0' to cancel all of the limit monitoring for the VID. Set 'p = 0' to cancel the specific LIMITID.

S2F46 Variable Limit Attributes Acknowledge (VLAA) M, H<-E
Description: Equipment acknowledges Host's limit monitoring.
Structure: L,2

```
    <VLAACK>
    L,m m = number of invalid parameters
        L,3
            <VIDp>VIDp = VID with error
            <LVACKp>LVACKp = reason
            L,n
                <LIMITID1>
                <LIMITACK1>
        …
        L,3
            <VIDx>
            <LVACKx>
            L,n
                <LIMITID1>
                <LIMITACK1>
```

Note: 'm=0' indicates no invalid limit attributes. 'n=0' indicates no invalid limit attributes for that VID.

S2F47 Variable Limit Attributes Request (VLAR) S, H->E, reply
Description: Host queries Equipment for current variable limit attribute definitions.
Structure: L,m

```
    <VID1>
    <VID2>
    …
    <VIDm>
```

Note: 'm' indicates the number of VIDs of which limit attributes are queried. Set 'm=0' to specify all VIDs which have limit attributes.

S2F48 Variable Limit Attributes Send (VLAS) M, H<-E
Description: Equipment sends Host values of requested variable limit attribute definitions in the order requested.
Structure: L,m
```
L,2
    <VID>
    L,p {p=0,4}
        <UNITS1>
        <LIMITMIN1>
        <LIMITMAX1>
        L,n
            L,3
                <LIMITID1>
                <UPPERDB1>
                <LOWERDB1>
            …
            L,3
                <LIMITIDn>
                <UPPERDBn>
                <LOWERDBn>
…
L,2
    <VIDm>
    L,p {p=0,4}
        <UNITS1>
        <LIMITMIN1>
        <LIMITMAX1>
        L,n
            L,3
                <LIMITID1>
                <UPPERDB1>
                <LOWERDB1>
            …
            L,3
                <LIMITIDn>
                <UPPERDBn>
                <LOWERDBn>
```
Note: 'm' is the number of VIDs in this request. 'n' is the number of LIMITIDs set on the VID. 'p=0' indicates that specific VID does not support limit monitoring function. 'n=0' indicates that no limit is set on the VID.

S2F49 Enhanced Remote Command (ERC) M, H<-E
Description: The host requests an object of perform the specified remote command with its associated parameters. If multi-block, it shall be preceded by the S2F39/F40 Multi-Block Inquire/Grant transaction.
Structure: L,4
```
<DATAID>
<OBJSPEC>
<RCMD>
```

```
L,m # of parameter groups
    L, 2
        <CPNAME1> command parameter 1 name
        <CEPVAL1> command enhanced parameter 1 value
    L, 2
        <CPNAME2> command parameter 2 name
        <CEPVAL2> command enhanced parameter 2 value
    …
    L, 2
        <CPNAMEm> command parameter m name
        <CEPVALm> command enhanced parameter m value
```

If a specific value of CPNAME is defined to have a CEPVAL defined as a LIST, it shall always be a LIST. If the CEPVAL that is associated to that specific value of CPNAME is defined to be anything other than LIST, it will result in a format error.

Exception: A zero length list, m=0, indicates that no parameter groups are sent with the command. OBJSPEC can get a null length item.

Note: If CEPVAL is a LIST, the items that make up a list of items with an identical format, as illustrated below.

```
<CPNAME>
L,m
    <CPVALa1>
    <CPVALa2>
    …
    <CPVALam>
```

S2F50 Enhanced Remote Command Acknowledge (ERCA) M, H<-E
Description: This equipment acknowledges Enhanced Remote Command or reports any error(s). If the command is not accepted due to one or more invalid parameters, (i.e. HCACK = 3), then a list of invalid parameters will be returned containing the parameter name and reason for being invalid
Structure: L, 2

```
<HCACK>
L, n # of parameters groups
    L, 2
        <CPNAME1>
        <CEPACK1>
    …
    L, 2
        <CPNAMEn>
        <CEPACKn>
```

4.3 Stream 5: Exception Reporting

This stream contains messages regarding equipment alarms. The alarms are generated by the equipment in response to changing conditions detected by the equipment.

S5F0 Abort Transaction (S5F0)
Description: Same form as S1F0.

S5F1 Alarm Report Send (ARS) S, H<-E, reply
Description: This message reports a change in or presence of an alarm condition. One message will be issued when the alarm is set and one when the alarm is cleared.
Structure: L,3
<ALCD>
<ALID>
<ALTX>

S5F2 Alarm Report Acknowledge (ARA) S, H->E
Description: Acknowledge or error.
Structure: <ACKC5>

S5F3 Enable/Disable Alarm Send (EAS) S, H->E, reply
Description: This message will change the state of the enable bit of all alarms in the equipment. The enable bit determines if the alarms will be sent to the host.
Structure: L,2
<ALED>
<ALID>
Note: A zero-length item for ALID means that all alarms are enabled or disabled.

S5F4 Enable/Disable Alarm Acknowledge (EAA) S, H<-E
Description: Acknowledge or error.
Structure: <ACKC5>

S5F5 List Alarms Request (LAR) S, H->E, reply
Description: This message requests the equipment to send alarm information to the host.
Structure: <ALID1,…ALIDn>
Exception: A zero length item means send all possible alarms regardless of the state of ALED.

S5F6 List Alarm Data (LAD) M, H<-E
Description: This message contains the alarm data known to the equipment. There are "m" alarms in the list.
Structure: L,m
L,3
<ALCD1>
<ALID1>
<ALTX1>
…
L,3

<ALCDm>
<ALIDm>
<ALTXm>
Exception: A zero-length item returned for ALTXi means that the given ALIDi does not exist.

S5F7 List Enabled Alarm Request (LEAR) S, H->E, reply
Description: This message requests the equipment to send enabled alarm information to the host.
Structure: Header only.

S5F8 List Enabled Alarm Data (LEAD) M, H<-E
Description: This message contains the enabled alarm data known to the equipment. There are "m" alarms in the list.
Structure: L,m
L,3
<ALCD1>
<ALID1>
<ALTX1>
…
L,3
<ALCDm>
<ALIDm>
<ALTXm>
Exception: A zero-length item returned for ALTXi means that enabled ALIDi does not exist.

4.4 Stream 6: Data Collection

This message stream covers the needs of in-process measurement and equipment monitoring.

S6F0 Abort Transaction (S6F0) S, H<—E
Description: Same form as S1F0.

S6F1 Trace Data Send (TDS) M, H<-E, reply
Description: Equipment reports trace data as specified in the S2F23.
Structure: L,4

```
<TRID>
<SMPLN>
<STIME>
L,n
    <SV1>
    <SV2>
    …
    <SVn>
```

Note: The SV's are arranged in the sequence of sampling cycle, and the data of the same sampling cycle are arranged in the order of the SVID's specified in the S2F23.

S6F2 Trace Data Acknowledge (TDA) S, H->E
Description: Host acknowledges trace data report.
Structure: <ACKC6>

S6F5 Multi-block Data Send Inquire (MBI) S, H<-E, reply
Description: If the discrete data report S6F11 can involve more than one block, this transaction precedes the transmission.
Structure: L,2

```
<DATAID>
<DATALENGTH>
```

S6F6 Multi-block Grant (MBG) S, H->E
Description: Grant permission to send multi-block message.
Structure: <GRANT>

S6F11 Event Report Send (ERS) S, H<-E, reply
Description: The purpose of this message is for the equipment to send a defined, linked and enabled group of reports to the host upon the occurrence of an event (CEID).
Structure: L,3

```
<DATAID>
<CEID>
L,a
   L,2 data of report #1
    <RPTID1>
        L,b
        <V1>
           ...
        <Vb>
```

...
L,2 data of report #a
<RPTIDa>
L,c
<V1>
...
<Vc>

Exception: If there are no reports linked to the event, a ‘null’ report is assumed. A zero-length list for the number of reports means there are no reports linked to the given CEID.

S6F12 Event Report Acknowledge (ERA) S, H->E
Description: Acknowledge or error.
Structure: <ACKC6>

S6F15 Event Report Request (ERR) S, H->E, reply
Description: Host requests the equipment to send an event report group by specifying its CEID.
Structure: <CEID>

S6F16 Event Report Data (ERD) M, H<-E
Description: Equipment sends host the reports linked to the CEID.
Structure: Same as S6F11.

S6F19 Individual Report Request (IRR) S, H->E, reply
Description: Host requests the equipment a defined report by specifying the report ID.
Structure: <RPTID>

S6F20 Individual Report Data (IRD) M, H<-E
Description: Equipment sends host variable data defined for the given RPTID.
Structure: L,n
<V1>
…
<Vn>

Note: The 'n=0' means RPTID is not defined.

S6F23 Request Spooled Data (RSD) S, H->E, reply
Description: Host requests the equipment to send spooled data.
Structure: <RSDC>

S6F24 Request Spooled Data Ack. Send (RSDAS) S, H<-E
Description: Equipment acknowledges spooled data send request and starts sending it.
Structure: <RSDA>

4.5 Stream 7: Process Program Management

The functions in this stream are used to manage and transfer process programs. Process programs are the equipment-specific descriptions that determine the procedure to be conducted on the material by a single equipment. Methods are provided to transfer programs between equipment and host.

S7F0 Abort Transaction (S7F0) S, H<—E
Description: Same form as S1F0.

S7F1 Process Program Load Inquire (PPI) S, H<->E, reply
Description: This message is used to initiate the transfer of a process program (recipe) or to select one from a number of stored programs. The message may be used to initiate the transfer of an unformatted process program (S7F3/F4).
Structure: L,2
<PPID>
<LENGTH>

S7F2 Process Program Load Grant (PPG) S, H<->E
Description: This message gives permission for the process program to be loaded.
Structure: <PPGNT>

S7F3 Process Program Send (PPS) M, H<->E, reply
Description: The program is sent.
Structure: L,2
<PPID>
<PPBODY>

S7F4 Process Program Acknowledge (PPA) S, H<->E
Description: Acknowledge or error.
Structure: <ACKC7>

S7F5 Process Program Request (PPR) S, H<->E, reply
Description: This message is used to request the transfer of a process program.
Structure: <PPID>

S7F6 Process Program Data (PPD) M, H<->E
Description: This message is used to request the transfer a process program.
Structure: L,2
<PPID>
<PPBODY>
Exception: A zero length list means request denied.

S7F17 Delete Process Program Send (DPS) S, H<->E, reply
Description: This message is used by the host to request the equipment deletes programs from the equipment storage (hard disk).
Structure: L,n (number of process programs to be deleted)
<PPID1>
…

<PPIDn>
Exception: If n = 0, then delete all process programs

S7F18 Delete Process Program Acknowledge (DPA) S, H<->E
Description: Acknowledge or error.
Structure: <ACKC7>

S7F19 Current EPPD Request (RER) S, H->E, reply
Description: This message is used to request the transmission of the current equipment process program directory (EPPD). This directory is a list of all the PPIDs of the process programs currently stored on the hard disk of the Equipment.
Structure: Header only

S7F20 Current EPPD Data (RED) S, H<-E
Description: This message is used to transmit the current equipment process program directory (EPPD).
Structure: L,n (number of process programs in the directory)
<PPID1>
…
<PPIDn>

4.6 Stream 9: System Errors

This message stream provides a method of informing the host that a message block has been received which cannot be handled, or that a timeout on a transaction (receive) timer has occurred. The message indicates that either a Message Fault or a Communications Fault has occurred, it does not indicate a Communications Failure.

S9F0 Abort Transaction (S9F0) S, H<—E
Description: Same form as S1F0.

S9F1 Unrecognized Device ID (UDN) S, H<-E
Description: The device ID in the message block header did not correspond to any known device ID on the equipment.
Structure: <MHEAD>

S9F3 Unrecognized Stream Type (USN) S, H<-E
Description: The equipment does not recognize the stream type in the message block header.
Structure: <MHEAD>

S9F5 Unrecognized Function Type (UFN) S, H<-E
Description: The equipment does not recognize the function type in the message block header.
Structure: <MHEAD>

S9F7 Illegal Data (IDN) S, H<-E
Description: This message indicates that the stream and function of the received message were recognized, but the associated data format could not be interpreted.
Structure: <MHEAD>

S9F9 Transaction Timer Timeout (TTN) S, H<-E
Description: This message indicates that a transaction (receive) timer has timed out and that the corresponding transaction has been aborted. It is up to the host to respond to this error in an appropriate manner to keep the system operational.
Structure: <SHEAD>

S9F11 Data Too Long (DLN) S, H<-E
Description: This message to the host indicates that the equipment has received more data than it can handle.
Structure: <MHEAD>

S9F13 Conversation Timeout (CTN) S, H<-E
Description: This message indicates that the equipment has expected data, but none was received within a reasonable length of time. In such cases, resources are cleared.
Structure: L,2
<MEXP>
<EDID>

4.7 Stream 10: Terminal Services

The function of this stream is to pass text messages between the host and the operator's monitor screen on the equipment. The equipment makes no attempt to interpret the text of the message, but merely displays it. Management of the human response, times etc, to the information displayed is the responsibility of the host.

S10F0 Abort Transaction (S10F0) S, H<—E
Description: Same form as S1F0.

S10F1 Terminal Request (TRN) S, H<-E, reply
Description: A terminal text message from the equipment to the host.
Structure: L,2
<TID>
<TEXT>

S10F2 Terminal Request Acknowledge (TRA) S, H->E
Description: Acknowledge or error.
Structure: <ACKC10>

S10F3 Terminal Display, Single (VTN) S, H->E, reply
Description: Data from the host to be displayed on the equipment.
Structure: L,2
<TID>
<TEXT>

S10F4 Terminal Display, Single Acknowledge (VTA) S, H<-E
Description: Acknowledge or error.
Structure: <ACKC10>

S10F5 Terminal Display, Multi-Block (VTN) M, H->E, reply
Description: Data from host to be displayed on the equipment monitor.
Structure: L,2
<TID>
L,n n = number of lines to display, maximum 8
<TEXT1>
…
<TEXTn>

S10F6 Terminal Display, Multi-Block Acknowledge (VTA) S, H<-E
Description: Acknowledge or error.
Structure: <ACKC10>

5 SECS SCENARIO

5.1 Alarm Management

5.1.1 Enable/Disable Alarms

Step	Host	Equipment	Description
1	S5F3 -->		The host requests that the alarm specified be enabled or disabled.
2		<-- S5F4	Equipment acknowledges with ACKC5 set to 0 if request was accepted, >0 if request was denied.

5.1.2 Send Alarm Report

Step	Host	Equipment	Description
1			The Equipment recognizes that an alarm has occurred. If reporting for this alarm ID is disabled, this scenario ends here.
2		<-- S5F1	Send the alarm report. The Equipment Constant WBitS5 determines whether the W-bit is 0 or 1 in this message.
3	S5F2 -->		If the W-bit in S5F1 is 1, the Host acknowledges the alarm report. Otherwise, skip this step.

5.1.3 Upload Alarm Information

Step	Host	Equipment	Description
1	S5F5 -->		The host requests the data and text for the specified alarms.
2		<-- S5F6	Equipment responds with the requested alarm data.

5.2 Clock

5.2.1 Equipment Requests Date and Time

Step	Host	Equipment	Description
1		<-- S2F17	Equipment requests current date and time from the Host.
2	S2F18 -->		Host sends its current date and time.

5.2.2 Host Instructs Equipment to Set Date and Time

Step	Host	Equipment	Description
1	S2F31 -->		Host instructs equipment to set date and time to the specified value.
2		<-- S2F32	Equipment sets internal time and acknowledges hosts request.

5.2.3 Host Requests Date and Time

Step	Host	Equipment	Description
1	S2F17 -->		Host requests current date and time from the equipment.
2		<-- S2F18	Equipment sends its current date and time.

5.3 ControlState

5.3.1 Equipment Initiated Are You There

Step	Host	Equipment	Description
1		<-- S1F1	Equipment sends Are You There periodically to determine if the SECS link is still intact.
2	S1F2 -->		Host replies with On Line Data. The Equipment knows that the link is still intact.

5.3.2 Host Sends Off Line

Step	Host	Equipment	Description
1	S1F15 -->		Host requests OFFLINE.
2			If equipment is ONLINE
3		<-- S1F16	Equipment acknowledges the request and transitions to OFFLINE, HOST OFFLINE state.
4		<-- S6F11	Equipment off-line event is sent.
5	S6F12 -->		Host acknowledges event.

5.3.3 Host Sends On Line

Step	Host	Equipment	Description
1	S1F17 -->		Host requests ONLINE.
2			If equipment is in state OFFLINE, HOST OFFLINE, go to step 4.
3		<-- S1F18	Equipment denies request (ONLACK > 0). Scenario ends here.
4		<--S1F18	Equipment acknowledges (ONLACK = 0) and goes ONLINE.
5		<-- S6F11	ControlStateLOCAL (or REMOTE) event.
6	S6F12 -->		Host acknowledges control state change.

5.3.4 Operator Sets Off-Line When Equipment In On-Line State

Step	Host	Equipment	Description
1			Operator actuates OFF-LINE switch on operator console when equipment ON-LINE state is active.
2		<-- S6F11	The Equipment sends "Equipment Requests OFF-LINE" event report (GemEquipmentOFFLINE). Event Reports as appropriate.
3	S6F12 -->		Host acknowledges the report.

5.3.5 Operator Sets On-Line (Remote) When Equipment In Off-Line State

Step	Host	Equipment	Description
1			Operator actuates REMOTE switch on operator console when equipment state is ONLINE/LOCAL.
2		<-- S6F11	The Equipment sends "ControlStateREMOTE" event report.
3	S6F12 -->		Host acknowledges the event.

5.3.6 Operator Sets On-Line (Local) When Equipment In Off-Line State

Step	Host	Equipment	Description
1			Operator actuates LOCAL switch on operator console when equipment state is ONLINE/REMOTE.
2		<-- S6F11	The Equipment sends “ControlStateLOCAL” event report.
3	S6F12 -->		Host acknowledges the event.

5.4 Data Collection

5.4.1 Equipment Reports Event

Step	Host	Equipment	Description
1			Equipment recognizes that an event has occurred. The Host has enabled reporting for the CEID, and possibly has defined one or more reports and linked them to the CEID. The equipment constant RpType is set to FALSE, requesting normal reports.
2			If event report is single block, go to step 5.
3		<-- S6F5	Inquire. If the event report is multi-block, the Equipment first sends this multi-block inquire for permission.
4	S6F6 -->		Grant. The Host grants permission to send multi-block event report. If GRANT6 is non-zero, this scenario fails here, and the event data is discarded.
5		<-- S6F11	Equipment sends event reports for the CEID that occurred.
6	S6F12 -->		Host acknowledges the report.

5.4.2 Event Reporting Setup

Step	Host	Equipment	Description
1			If the define report message is single block, go to step 4.
2	S2F39 -->		Inquire. If the define report is multi-block, the host first sends this multi-block inquire for permission.
3		<-- S2F40	Grant. The equipment grants permission to send multi-block report definition. If GRANT is non-zero, this scenario fails here.
4	S2F33 -->		Define report. The host sends report definitions.
5		<-- S2F34	The equipment acknowledges. If the reports are accepted, DRACK=0.
6			If the link event report message is single block, go to step 9.
7	S2F39 -->		Inquire. If the link event report is multi-block, the host first sends this multi-block inquire for permission.
8		<-- S2F40	Grant. The equipment grants permission to send multi-block link event report. If GRANT is non-zero, this scenario fails here.
9	S2F35 -->		Link Events/Reports. The host links reports to the desired collection events. Linked reports are initially “disabled”.
10		<-- S2F36	Equipment acknowledges. If LRACK is zero, the Event/Report linkages are acceptable.
11	S2F37 -->		Enable Event Reports. The host enables reporting for the desired collection events.
12		<-- S2F38	Equipment acknowledges. If ERACK is zero, the equipment will report events as they occur.

5.4.3 Host Requests Equipment Constants

Step	Host	Equipment	Description
1	S2F13 -->		The host requests the value of one or more equipment constants.
2		<-- S2F14	Equipment sends the values of the requested constants.

5.4.4 Host Requests Equipment Constants Namelist

Step	Host	Equipment	Description
1	S2F29 -->		The host requests information about the specified equipment constants (e.g. name, min, max, default, units.)
2		<-- S2F30	Equipment sends the requested data.

5.4.5 Host Requests Event Report

Step	Host	Equipment	Description
1	S6F15 -->		The host requests an event report group by specifying the CEID.
2		<-- S6F16	The equipment sends any reports linked to the requested collection event.

5.4.6 Host Requests Report (Variable Data Collection)

Step	Host	Equipment	Description
1	S6F19 -->		The host requests an individual report request
2		<-- S6F20	The equipment sends the requested report data.

5.4.7 Host Requests Status Variable Namelist

Step	Host	Equipment	Description
1	S1F11 -->		The host requests that the equipment identify the specified status variables.
2		<-- S1F12	Equipment responds with the requested status variable data.

5.4.8 Host Request Status Variable Report

Step	Host	Equipment	Description
1	S1F3 -->		The host requests the VIDs of interest.
2		<-- S1F4	Equipment responds with the requested status variable values.

5.4.9 Set Equipment Constants

Step	Host	Equipment	Description
1	S2F15 -->		The host requests to change the values of one or more equipment constants.
2		<-- S2F16	Equipment acknowledges. If EAC is zero, the ECs have been modified successfully.

5.4.10 Trace Data Collection

Step	Host	Equipment	Description
1	S2F23 -->		The host initiates a trace.
2		<-- S2F24	The equipment acknowledges the trace request. If the data in S2F23 is not valid, TIAACK is non-zero and the scenario ends. Otherwise, the following are done "TOTSMP" times, where TOTSMP is the total number of samples to be done.
3			Collect data.
4			Wait DSPER.
5			If number of samples < REPGSZ, go to step 3.
6		<-- S6F1	The equipment sends the trace data.
7	S6F2 -->		If the S6F1 has its W-Bit set to 1, the host acknowledges the trace data.
8			If number of samples sent is less than TOTSMP, go to step 3.
9	S2F23 -->		[Optional] Host requests trace termination prior to completion by initiating a trace with the same trace ID as the currently running trace and with TOTSMP set to "0".
10		<-- S2F24	The equipment acknowledges the trace request. If the data in S2F23 is valid, the equipment terminates the trace. If the equipment has saved trace data that has not yet been sent to the host, it discards the saved data.

5.5 Error Messages

5.5.1 Message Fault due to Unrecognized Device ID

Step	Host	Equipment	Description
1	SxFy -->		Host sends a message with a bad Device ID in the header.
2		<-- S9F1	Equipment replies with "Unrecognized Device ID".

5.5.2 Message Fault due to Unrecognized Stream

Step	Host	Equipment	Description
1	SxFy -->		Host sends primary message with a stream number that the Equipment does not support.
2		<-- S9F3	Equipment replies with "Unrecognized Stream".

5.5.3 Message Fault due to Unrecognized Function

Step	Host	Equipment	Description
1	SxFy -->		Host sends primary message with a stream number that the Equipment supports, but with a function number that the Equipment does not support for that stream.
2		<-- S9F5	Equipment replies with "Unrecognized Function".

5.5.4 Message Fault due to Illegal Data Format

Step	Host	Equipment	Description
1	SxFy -->		Host sends a message with a stream and function that the Equipment recognizes, but with a data format that is incorrect.
2		<-- S9F7	Equipment replies with "Illegal Data Format".

5.5.5 Message Fault due to Transaction Timer Time-out

Step	Host	Equipment	Description
1			Equipment does not receive an expected reply message from the host and a transaction time-out occurs.
2		<-- S9F9	Equipment sends "Transaction Timer Time-out".

5.5.6 Message Fault due to Data Too Long

Step	Host	Equipment	Description
1	SxFy -->		Host sends a message with a stream and function that the Equipment recognizes, but contains more data than expected.
2		<-- S9F11	Equipment replies with "Data Too Long". If the erroneous message is a primary message with the W-bit set to 1, then in some cases the Equipment will reply with the usual secondary response with an appropriate error code, instead of S9F11. If the erroneous message is secondary, the Equipment makes no reply at all.

5.5.7 Message Fault due to Conversation Timeout

Step	Host	Equipment	Description
1	SxFy -->		Host sends a message with a stream and function that the Equipment recognizes,
2		<-- SxFy +1	
3		<-- S9F13	Conversation timeout occurred before Equipment received an expected message from Host. Note : The equipment will detect a conversation timeout only between the receipt of an S7F1/F2 and an S7F3 from Host.

5.6 Establish Communications

5.6.1 Equipment Attempts to Establish Communications

Step	Host	Equipment	Description
1			The equipment communication state is NOT COMMUNICATING.
2		<-- S1F13	Equipment sends Establishes Communications Request.
3	S1F14 -->		Host sends Establish Communications Acknowledge.
4			If S1F14 received without time-out, then go to step 7.
5			Wait for EstablishCommunicationsTimeout period.
6			Go to step 2.
7			If COMMACK = Accept(0), then go to step 10.
8			Wait for EstablishCommunicationsTimeout period.
9			Go to step 2.
10			Communications established.

5.6.2 Equipment Attempts to Establish Communications Denied

Step	Host	Equipment	Description
1			The equipment communication state is ENABLED.
2	S1F13 -->		Host sends Establishes Communications Request.
3		<-- S1F14	The equipment responds with Establish Communications Acknowledge, with COMMACK ≠ Accept (0).
4			Communications not established.

5.6.3 Equipment Attempts to Establish Communications Fails

Step	Host	Equipment	Description
1			The equipment communication state is NOT COMMUNICATING.
2		<-- S1F13	Equipment sends Establishes Communications Request.
3			Wait for EstablishCommunicationsTimeout period.
4			If S1F14 received with time-out, then go to step 5
5			Communications not established.

5.6.4 Host Attempts to Establish Communications

Step	Host	Equipment	Description
1			The equipment communication state is ENABLED.
2	S1F13 -->		Host sends Establishes Communications Request.
3		<-- S1F14	The equipment responds with Establish Communications Acknowledge, with COMMACK = Accept (0).
4			Communications established.

5.6.5 Host Initiated Are You There

Step	Host	Equipment	Description
1	S1F1 -->		Host sends Are You There.
2		<-- S1F2	Equipment replies with its MDLN and SOFTREV.

5.6.6 Simultaneous Attempt to Establish Communications (I)

The equipment receives S1F14 from the host before sending S1F14.

Step	Host	Equipment	Description
1			The equipment communication state is NOT COMMUNICATING.
2		<-- S1F13	Equipment sends Establish Communications Request.
3	S1F13 -->		Host sends Establish Communications Request.
4	S1F14 -->		Host sends Establish Communications Acknowledge with COMMACK = Accept (0).
5			At this point, communications are established.
6		<-- S1F14	Equipment sends S1F14 with COMMACK = Accept (0).

5.6.7 Simultaneous Attempt to Establish Communications (II)

The equipment sends S1F14 to the host before receiving S1F14.

Step	Host	Equipment	Description
1			The equipment communication state is NOT COMMUNICATING.
2		<-- S1F13	Equipment sends Establish Communications Request.
3	S1F13 -->		Host sends Establish Communications Request.
4		<-- S1F14	Equipment sends S1F14 with COMMACK = Accept (0).
5			At this point, communications are established.
6	S1F14 -->		Host sends Establish Communications Acknowledge with COMMACK = Accept (0).

5.7 Limit Monitoring

5.7.1 Host Defines Limit Attributes

Step	Host	Equipment	Description
1			If the define report message is single block, go to step 4.
2	S2F39 -->		Inquire. If the define report is multi-block, the host first sends this multi-block inquire for permission.
3		<-- S2F40	Grant. The equipment grants permission to send multi-block report definition. If GRANT is non-zero, this scenario fails here.
4	S2F45 -->		Host defines new variable limit attributes.
5		<-- S2F46	Equipment acknowledges Host request.

5.7.2 Host Queries Equipment for Current Limits

Step	Host	Equipment	Description
1	S2F47 -->		Host queries Equipment current variable limit attribute definitions
2		<-- S2F48	Equipment returns report containing requested variable limit attribute values.

5.8 Process Program Management

5.8.1 Program Deletion by the Host

Step	Host	Equipment	Description
1	S7F17 -->		Host requests deletion of a process program.
2		<-- S7F18	Equipment acknowledges or reports error.

5.8.2 Program Directory Request

Step	Host	Equipment	Description
1	S7F19 -->		The host requests a list of process programs available in the equipment.
2		<-- S7F20	Equipment sends the requested data.

5.8.3 Host-initiated Program Download Request

Step	Host	Equipment	Description
1	S7F1 -->		The host requests permission to send a process program to the equipment.
2		<-- S7F2	The equipment grants or denies permission for the host to send the process program. If the value of the grant code is non-zero, permission is denied and this scenario ends here.
3	S7F3 -->		The host sends the process program to the equipment.
4			Equipment validates the process program.
5		<-- S7F4	The equipment acknowledges receipt of the process program. If the process program is valid the acknowledge code is zero. If the process program is not valid, or could not be stored in the library, the acknowledge code in this message is non-zero.

5.8.4 Host-initiated Process Program Upload (Unformatted)

Step	Host	Equipment	Description
1	S7F5-->		The host requests that the equipment send the specified process program.
2		<-- S7F6	Equipment sends the requested data.

5.9 Remote Control

5.9.1 Host Sends Remote Command to Equipment

Step	Host	Equipment	Description
1	S2F41 -->		Host sends command.
2		<-- S2F42	The equipment acknowledges the command

5.9.2 Host Sends Enhanced Remote Command to Equipment

Step	Host	Equipment	Description
1	S2F49 -->		Host sends the enhanced command
2		<-- S2F50	Equipment acknowledges the enhanced command

5.10 Spooling

5.10.1 Define Spooled Messages

Step	Host	Equipment	Description
1	S2F43 -->		Host specifies the streams and functions to be spooled in case of communications failure.
2		<-- S2F44	Equipment acknowledges setup.

5.10.2 Request or Delete Spooled Data

Step	Host	Equipment	Description
1			Communications between the host and equipment fail.
2			Communications between the host and equipment are restored.
3	S1F3 -->		Host requests variable data that includes spool related status variables. Spool variables of interest are SpoolCountActual, SpoolCountTotal, SpoolFullTime, and SpoolStartTime. Note: Any valid method of obtaining status variable data may be used in place of S1F3.
4		<-- S1F4	The equipment sends the status data.
5	S6F23 -->		Host sends request for spooled data or delete of spooled data depending on value of RSDC. RSDC = 0 spool data is requested RSDC = 1 spool data is to be discarded
6		<-- S6F24	Equipment acknowledges request.
7			If RSDC = 1 (delete), go to step 11.
8		<-- SxFy	If RSDC is set to 0, send spooled data. If MaxSpoolTransmit = 0, then go to step 11. If MaxSpoolTransmit > 0, the MaxSpoolTransmit oldest messages are transmitted to the Host. Spooling remains active.
9			If spool is empty, go to step 11.
10	S6F23 -->		Host recognizes that MaxSpoolTransmit is reached. Host sends request for additional spooled data (RSDC = 0). Go to step 6.
11		<-- S6F11	The Equipment sends Spooling Deactivated event report (GemSpoolingDeactivated). Event reports as appropriate.
12	S6F12 -->		Host acknowledges the report.

5.11 Terminal Service

5.11.1 Host Sends Information to Equipment’s Display Device

Step	Host	Equipment	Description
1	S10F3 -->		Host sends textual information to the equipment terminal.
2		<-- S10F4	Equipment acknowledges receipt of the text. Note that this only acknowledges that the text was received, not that the operator has seen it yet.
3			Operator acknowledges the text by pressing the OK button on the operator’s console.

5.11.2 Host Sends Multi-block Text Message to Equipment

Step	Host	Equipment	Description
1	S10F5 -->		Host sends multiple blocks of textual information to the equipment terminal.
2		<-- S10F6	Equipment acknowledges receipt of the text. Note that this only acknowledges that the text was received, not that the operator has seen it yet.
3			Operator acknowledges the text by pressing the OK button on the operators console.

5.11.3 Operator Sends Information to the Host

Step	Host	Equipment	Description
1			Operator enters textual information at the equipment operator’s console and presses the SEND button.
2		<-- S10F1	Equipment sends the text to the host.
3	S10F2 -->		Equipment acknowledges the text.

6 REMOTE COMMANDS

Table 6-1 lists the remote commands available on the system. The format for each command is given in Table 6-2, and parameters used with each command are listed in Table 6-3.

Table 6-1: Remote Commands

RCMD	Description
SETUPJOB	Job Information received to be store in database by equipment
STARTJOB	Start Job information received as permission to start a job

Table 6-2: Remote Command Formats

RCMD and Format	Description
SETUPJOB L, 2 1. “SETUPJOB” 2. L, 3 1. L, 2 1. “JobParameters” 2. L, 2 1. L, 2 1. “JobName” 2. <JobName> 2. L, 2 1. “LotNumber” 2. <LotNumber> 2. L, 2 1 “RecipeParameter” 2. L, 2 1. L, 2 1. “CT” 2. <CT> 2. L, 2 1. “ST” 2. <ST> 3. L, 2 1. “Carriers” 2. L, n 1. <Carriers>	JobParameters consist of 2 elements which are <JobName> and <LotNumber> RecipeParameter consist of 2 elements which are <CT> and <ST> Carrier consist a list of elements which are <Carriers>
STARTJOB L, 2 1. “JobName” 2. <JobName>	JobName consist of an element which is <JobName>

Table 6-3: Remote Command Parameters

CPNAME	Used In	Description	Format
JobName	SETUPJOB	JobName to be run by equipment	A, 20 bytes
LotNumber	SETUPJOB	LotNumber to be run by equipment	A, 20 bytes
CT	SETUPJOB	Cooling Time in seconds	A, 20 bytes

ST	SETUPJOB	Staging Time in seconds	A, 20 bytes
Carriers	SETUPJOB	List of carriers ID	A, 20 bytes
JobName	STARTJOB	JobName to be start by equipment	A, 20 bytes

APPENDIX A: DATA ITEM DICTIONARY

The following data items are used by the SECS-II messages (SECS-II Messages Detail).

Data Item	Description/ Codes	Format	Where Used
ABS	Any binary string to confirm communication line is alive. Variable data length, maximum of 40 bytes.	B	S2F25,F26
ACKC5	Acknowledge code, 1 byte 0 = Accepted 64 = ALED has an illegal value 65 = ALID has an illegal value * ACKC5 not = 0 returned from HOST is regarded as accepted.	B	S5F2,F4
ACKC6	Acknowledge code, 1 byte 0 = Accepted and done >0 = Error, not done * ACKC6 not = 0 returned from HOST is regarded as accepted.	B	S6F12
ACKC7	Acknowledge code, 1 byte 0 = Accepted 1 = Permission not granted 2 = Length error 3 = Matrix overflow 4 = PPID (recipe file) not found 64 = Illegal PPID (recipe file name)	B	S7F4,F18
ACKC10	Acknowledge code, 1 byte 0 = Accepted for display 1 = Message will not be displayed 2 = Terminal not available	B	S10F2,F4,F6
ALCD	Alarm code, 1 byte bit 8 = 1 means alarm set bit 8 = 0 means all alarms cleared bit 7-1 are not used	B	S5F1,F6
ALED	Alarm enable/disable code, 1 byte bit 8 = 1 means enable all alarms bit 8 = 0 means disable all alarms	B	S5F3
ALID	Alarm identification	U4	S5F1,F3,F5,F6
ALTX	Alarm text	A	S5F1
CEED	Collection event enable/disable code, 1 byte FALSE = Disable TRUE = Enable	BL	S2F37
CEID	Collected Event ID	U4	S2F35,S6F11,F15,F16
CEPACK	Command enhanced parameter acknowledge, 1 byte 1 = Parameter Name (CPNAME) does not exist 2 = Illegal Value specified for CEPVAL	B	S2F50

	3 = Illegal Format specified for CEPVAL 4 = Parameter Name (CPNAME) not valid as used		
COMMACK	Establish communications acknowledge code, 1 byte 0 = Accepted. 1 = Denied, try again.	B	S1F14
CPACK	Command parameter acknowledge code, 1 byte 1 = Parameter Name (CPNAME) does not exist 2 = Illegal Value specified for CPVAL 3 = Illegal Format specified for CPVAL	B	S2F42
CPNAME	Command Parameter Name. User defined.	A	S2F41,F42,F49,F50
CPVAL	Command Parameter Value.	Any, except L	S2F41
DATAID	Data Identifier. A unique name used to link multi-block messages for a single event report.	U4	S2F33,F35, F39,F49,S6F5,F11,F16
DATALENGTH	Total number of bytes to be sent.	U4	S2F39,S6F5
DRACK	Acknowledge code, 1 byte 0 = Accepted. 1 = Denied. Insufficient space. 2 = Denied. Invalid format. 3 = Denied. At least one RPTID already defined. 4 = Denied. At least one VID does not exist.	B	S2F34
DSPER	Character string to indicate data sampling period, 6 bytes hhmmss	A	S2F23
EAC	Equipment acknowledge code, 1 byte 0 = Acknowledge 1 = Denied, at least one constant does not exist 2 = Denied, busy 3 = Denied, at least one constant out of range	B	S2F16
ECDEF	Equipment constant default value	Any	S2F30
ECID	Equipment constant ID	U4	S2F13,F15,F29,F30
ECMAX	Equipment constant maximum value	Any	S2F30
ECMIN	Equipment constant minimum value	Any	S2F30
ECNAME	Equipment constant name **Note:** A zero length value means that an illegal ECID has been given or that	A	S2F30

	the ECV doesn't exist with the actual configuration or settings of the equipment.		
ECV	Equipment constant value **Note:** A zero length value means that an illegal ECID has been given or that the ECV doesn't exist with the actual configuration or settings of the equipment.	Any	S2F14,F15
EDID	Expected data identification One possible response MEXP EDID S7F3 <PPID>	A	S9F13
ERACK	Enable/disable event report acknowledge code, 1 byte 0 = Accepted 1 = Denied. At least one CEID does not exist	B	S2F38
FCNID	Function number	U1	S2F43,F44
GRANT	Permission to send, 1 byte 0 = Permission granted 1 = Busy, try again 2 = Insufficient space	B	S2F40,S6F6
HCACK	Host command parameter acknowledge code, 1 byte 0 = Acknowledge, command has been performed 1 = Command does not exist 2 = Cannot perform now 3 = At least one parameter is invalid 4 = Acknowledge, command will be performed later 5 = rejected, not in remote mode / already in desired condition 6 = Invalid object	B	S2F42,F50
LENGTH	Length of the process program in bytes	U4	S7F1
LIMITACK	Acknowledge code for variable limit attribute set, 1 byte 1 = LIMITID does not exist 2 = UPPERDB > LIMITMAX 3 = LOWERDB < LIMITMIN 4 = UPPERDB < LOWERDB 5 = UPPERDB/LOWERDB format error 6 = ASCII character string cannot be translated into numeric 7 = Duplicated variable limit is set for this variable 64 = Error occurred on equipment software	B	S2F46
LIMITID	Identifier of Limit (as defined by UPPERDB and LOWERDB), 1 byte	B	S2F45,F46,F48

LIMITMAX	The maximum value allowed for the specific variable	Any	S2F48
LIMITMIN	The minimum value allowed for the specific variable	Any	S2F48
LOWERDB	The variable limit attribute, which defines the lower boundary of Limit ID. Specified together with UPPERDB	Any	S2F45,F48
LRACK	Acknowledge code, 1 byte 0 = Accepted. 1 = Denied. Insufficient space. 2 = Denied. Invalid format. 3 = Denied. At least one CEID link already defined. 4 = Denied. At least one CEID does not exist. 5 = Denied. At least one RPTID does not exist.	B	S2F36
LVACK	Variable limit definition acknowledge code, 1 byte. 1 = Variable does not exist 2 = Variable has no limit capability 3 = Same variable is repeated in the message 4 = Limit value error as described in LIMITACK 64 = Error occurred on equipment software	B	S2F46
MDLN	Equipment model	A	S1F2,F13,F14,S7F23,F26
MEXP	Message expected in the form SxFy, where x is stream and y is function	A	S9F13
MHEAD	SECS message block header, associated with message block in error, 10 bytes	B	S9F1,F3,F5,F7,F11
OBJSPEC	A text string that is used to point to a specific object instance	A	S2F49
OFLACK	Acknowledge code for OFF-LINE request, 1 byte 0 = OFF-LINE acknowledge	B	S1F16
ONLACK	Acknowledge code for ON-LINE request, 1 byte 0 = ON-LINE accepted 1 = ON-LINE not allowed 2 = Equipment already ON-LINE	B	S1F18
PPBODY	Process Program Body (recipe file)	B	S7F3,F6
PPGNT	Process Program Grant status, 1 byte 0 = OK 1 = Already have 2 = No space 3 = Invalid PPID 4 = Busy, try later 5 = Other error	B	S7F2

PPID	Process program ID (recipe file name)	A	S7F1,F3,F5,F6,F23,F25,F26
RCMD	Remote command code. User defined	A	S2F41,F49
REPGSZ	Reporting group size	U4	S2F23
RPTID	Report ID. The grouping ID of variable data. Report ID <= 100 is reserved for predefined RPTID in the equipment system. Host need to use RPTID > 100 to define report.	U4	S2F33,F35,S6F11,F16,F19
RSDA	Request spool data acknowledge code, 1 byte 0 = OK 1 = Denied, busy try again later 2 = Denied, spooled data does not exist 64 = Error occurred on equipment software	B	S6F24
RSDC	Request spool data code 0 = Transmit spooled messages 1 = Purge spooled message	U1	S6F23
RSPACK	Reset spooling acknowledge code, 1 byte 0 = Acknowledge, spooling setup accepted 1 = Spooling setup rejected 64 = Error occurred on equipment software	B	S2F44
SHEAD	Stored message header related to the transaction timer, 10 bytes	B	S9F9
SMPLN	Sample number. Indicates the sampling count of trace report. The first time is counted one, and increased by one	U4	S6F1
SOFTREV	Software Revision code	A	S1F2,F13,F14,S7F23,F26
STIME	Character string expressing latest sampled time, 12 bytes YYMMDDhhmmss	A	S6F1
STRACK	Spool set stream acknowledge code, 1 byte 1 = Specified Stream cannot be set (example: Stream = 1) 2 = Specified Stream is invalid 3 = Specified Stream/Function is invalid 4 = Secondary message was specified 64 = Error occurred on equipment software	B	S2F44
STRID	Stream number	U1	S2F43,F44
SV	Status variable value **Note:**	Any	S1F4,S6F1

	A zero length value means that an illegal SVID has been given or that the SV doesn't exist with the actual configuration or settings of the equipment.		
SVID	Status variable data ID	U4	S1F3,F11,F12,S2F23
SVNAME	Status variable name **Note:** A zero length value means that an illegal SVID has been given or that the Status Value doesn't exist with the actual configuration or settings of the equipment.	A	S1F12
TEXT	A single line of characters	A	S10F1,F3,F5
TIAACK	Trace acknowledge code, 1 byte 0 = Everything correct 1 = Too many SVIDs 2 = No more trace allowed 3 = Invalid sampling period 4 = Duplicated SVIDs 5 = SVIDs not exist 64 = Error occurred on equipment software	B	S2F24
TIACK	Time acknowledge code, 1 byte 0 = OK 1 = Error, not done	B	S2F32
TID	Terminal number, 1 byte. 0 = Main terminal	B	S10F1,F3,F5
TIME	Time of day. Depending upon the ECV "TimeFormat" the equipment sends the following formats: YYMMDDhhmmss 12 characters YYYYMMDDhhmmsscc 16 characters The equipment accepts both formats from the host	A	S2F18,F31
TOTSMP	Total samples to be made	U4	S2F23
TRID	Trace request ID	U4	S2F23,S6F1
UNITS	Units identifier of a SV or ECV. **Notes:** A zero length value means that no units are defined for the given SV or ECV (Example: any error counter). A zero length value will also be returned, if an illegal SVID or ECID has been given or if the SV or ECV doesn't exist with the actual configuration or settings of the equipment.	A	S1F1,S2F30,F48
UPPERDB	The variable limit attribute, which defines the upper boundary of Limit ID. Specified together with LOWERDB	Any	S2F45,F48

V	Variable data value (includes SVs, ECVs, DVVALs).	Any	S6F11,F16,F20
VID	Variable data ID (includes SVIDs, ECIDs, DVIDs).	U4	S2F33,F45,F46,F47,F48
VLAACK	Variable limit attribute acknowledge code, 1 byte 0 = Acknowledge 1 = Limit attribute definition error 2 = Cannot perform now 64 = Error occurred on equipment software	B	S2F46

APPENDIX B: VARIABLE ITEM DICTIONARY

This section describes variables and constants related specifically to the GEM interface itself.

1-100 = reserved for GEM Variables
Above 101 = reserved for Cooling Equipment Variables

Communications Data Items

VID	Type	Format	Description
1	EC	U1	**InitCommState** Determines whether communications are enabled or disabled following system initialization. 0 = Disabled 1 = Enabled (default)
2	SV	U1	**CommState** Indicates current communication state of the system. 0 = Disabled 1 = Enabled-Not Communicating 2 = Enabled-Communicating
3	EC	U2	**EstablishCommunicationTimeout** Determines how often the equipment attempts to establish communications with the host. Units = s. Valid range is 0 - 1800. Default is 30 seconds. A value of 0 means the equipment will not attempt to establish contact, but will wait for the host to do so.
4	EC	U2	**HeartBeat** Determines how long the equipment waits after sending the last message before sending an S1F1 “heartbeat” message. Units = s. Valid range is 0 - 1800. Default is 30 seconds. A value of 0 disables the heartbeat.

Command Data Items

VID	Type	Format	Description
5	SV	A	**PreviousCommand** Name of the last command issued to the equipment.

Alarm Data Items

VID	Type	Format	Description
6	EC	U1	**WBITS5** Determines whether S5 messages will have the “W” bit set (i.e., will expect a reply from the host). 0 = W bit cleared 1 = W bit set (default)
7	DV	U4	**AlarmID** Indicates the alarm ID of the cause of the latest transition.
8	SV	L	**AlarmsEnable** A list of the currently enabled alarms.
9	SV	L	**AlarmsSet** A list of the alarms currently in the set (or unsafe) condition.
10	SV	U1	**AlarmState** Indicates the direction of the latest alarm transition. 0 = ON to OFF 1 = OFF to ON

Event Reporting Data Items

VID	Type	Format	Description
11	SV	U4	**PreviousCEID** Indicates the CEID of the previous event.
12	EC	U1	**WBITS6** Determines whether S6 messages will have the “W” bit set (i.e., will expect a reply from the host). 0 = W bit cleared 1 = W bit set (default)
13	SV	L	**EventsEnabled** A list of events that are enabled.

Clock Data Items

VID	Type	Format	Description
14	SV	A	**Clock** The host’s view of the current time and date. Units = cs. The format of the data is YYYYMMDDhhmmsscc
15	EC	U1	**TimeFormat** Determines whether a 12-byte or 16-byte value (required for Year 2000 compliance) for the current date/time will be used. 0 = 12-byte format 1 = 16-byte format (default)

ControlState Data Items

VID	Type	Format	Description
16	EC	U1	**InitControlState** Determines whether the equipment comes up in the on-line or off-line control state. 1 = Off-line (default) 2 = On-line
17	EC	U1	**OfflineSubstate** Determines what off-line substate the equipment is to come up in if the InitControlState is set to Off-line. 1 = Off-line, equipment off-line (default) 2 = Off-line, attempt on-line 3 = Off-line, host off-line.
18	EC	U1	**OnlineFailed** Determines what off-line substate the equipment enters if an attempt to go on-line fails. 1 = Off-line, equipment off-line (default) 3 = Off-line, host off-line
19	EC	U1	**OnlineSubstate** Determines what on-line substate the equipment enters when going on-line. 4 = On-line, local (default) 5 = On-line, remote
20	SV	U1	**ControlState** 1 = Off-line, equipment off-line 2 = Off-line, attempt on-line 3 = Off-line, host off-line 4 = On-line, local 5 = On-line, remote
21	SV	U1	**PreviousControlState** 1 = Off-line, equipment off-line (default) 2 = Off-line, attempt on-line 3 = Off-line, host off-line 4 = On-line, local 5 = On-line, remote

Process Program Data Items

VID	Type	Format	Description
22	DV	A	**PPChangeName** The name of the last process program to be changed. Valid on GemPPChangeEvent.
23	DV	U1	**PPChangeStatus** Indicates how a process program has been changed. 0 = Unknown 1 = Created 2 = Changed 3 = Deleted
24	DV	A	**PPExecName** The name of the process program currently loaded into the equipment.
25	EC	A	**PPDirectory** Indicates the directory, where all the recipes stored.
26	EC	A	**PPFileExt** Indicates the recipe extension.

Limit Monitoring Data Items

VID	Type	Format	Description
27	DV	B	**EventLimit** Limit ID that triggered zone transition of limit monitoring function.
28	DV	U4	**LimitVariable** SVID that triggered zone transition by limit monitoring function.
29	DV	B	**TransitionType** Data indicates the direction of zone transition invoked by limit monitoring. 0 = Lower to Upper 1 = Upper to Lower

Spooling Data Items

VID	Type	Format	Description
30	EC	U4	**MaxSpoolTransmit** Determines how many messages will be transmitted to the host when the host requests the spool contents. A value of 0 indicates all messages will be sent. Default = 0.
31	EC	U1	**ConfigSpool** Determines whether spooling is enabled or disabled. 0 = Disabled (default) 1 = Enabled
32	SV	U4	**SpoolCountActual** The number of messages actually contained in the spool file.
33	SV	U4	**SpoolCountTotal** The number of messages that have been (or would have been) written to the spool since the last transmit or purge.
34	EC	BL	**OverWriteSpool** Indicates what to do when the spool file becomes full. TRUE = Overwrite oldest message (default) FALSE = Discard new messages
35	SV	A	**SpoolFullTime** Indicates the date and time that the spool became full. Units = cs. Data format is YYYYMMDDhhmmsscc.
36	SV	U1	**SpoolLoadSubstate** Indicates the current substate of the Spool Load state model. 6 = Spool not full 7 = Spool full
37	SV	A	**SpoolStartTime** Indicates the date and time that spooling started. Units = cs. Data format is YYYYMMDDhhmmsscc.
38	SV	U1	**SpoolState** Indicates the current state of the spooling state model. 1 = Spooling active 2 = Spooling inactive
39	SV	U1	**SpoolUnloadSubstate** Indicates the current state of the Spool Unload state model. 3 = Purge spool 4 = Transmit spool 5 = No spool output

Terminal Services Data Items

VID	Type	Format	Description
40	EC	U1	**WBITS10** Determines whether S10 messages from the equipment will have the W (wait for reply) bit set or cleared. 0 = W bit cleared 1 = W bit set (default)

Data Items

VID	Type	Format	Description
41	EC	A	**EquipmentName** Contains a description of the equipment.
42	SV	A	**PreviousProcessState** Indicates the previous processing state. See "ProcessState" for values.
43	SV	A	**ProcessState** Indicates the current processing state. Values are as follows: 0 = INIT 1 = IDLE 2 = RUN 3 = ERROR 4 = PURGING
44	SV	A	**EquipmentModel** Indicates the equipment model number (MDLN).
45	SV	A	**SoftwareRevision** Indicates the software revision (SOFTREV).

Cooling Equipment Data Items

VID	Type	Format	Description
101	DV	A	**Job ID** Indicates the Job ID. Example: JobID1
102	DV	A	**Carrier ID** Indicates the Carrier ID. Example: Carrier1
103	DV	A	**Load Port ID** Indicates the Load Port ID. Example: A
104	DV	A	**Job Status** Indicates the status for a Job Example: Queued, Selected, WaitingForStart, Executing, Completed and Destroyed
105	DV	L	**Carrier ID and Load Port ID to be load** Indicates the carrier to be load on which load port.
106	DV	L	**Job ID and Lot ID Status** Indicates the Job ID and Lot ID current status.
107	DV	L	**Carrier ID and Load Port ID to be unload** Indicates the carrier ready to unload
1050	SV	A	**Load Port ID and Carrier ID** Indicates the Load Port ID and Carrier ID to load

			Example: A,Carrier1
1060	SV	A	**Job ID, Job Status and Lot ID** Indicates the Job ID, Job Status and Lot ID Example: Job1,Executing,Lot1
1070	SV	A	**Load Port ID and Carrier ID** Indicates the Load Port ID and Carrier ID for unload Example: A,Carrier1

APPENDIX C: COLLECTION EVENT

As with the data items, there are events that are related to the functioning of the GEM interface and events that are related to the functioning of the Cooling Equipment system. This section is divided into <two> subsections along these lines.

1-100 = reserved for GEM Events
101-5000 = reserved for Cooling Equipment Events

GEM Events

CEID	Description/Trigger	Default RPTID
1	**GemControlStateLOCAL** Indicates that the GEM control state has transitioned to LOCAL control.	1
2	**GemControlStateREMOTE** Indicates that the GEM control state has transitioned to REMOTE control.	1
3	**GemEquipmentOFFLINE** Indicates that the GEM control state has transitioned to the OFFLINE state.	1
4	**GemPPChangeEvent** Indicates that a process program on the equipment has been changed. Suggested Data: PPChangeName, PPChangeStatus	2
5	**GemProcessStateChange** Indicates that a process state on the equipment has been changed.	3
6	**GemLimitZoneTransition** Indicates that limit zone transition has occurred.	4
7	**GemSpoolingActivated** Indicates that spooling has been activated and messages are being spooled. Suggested Data: SpoolStartTime	5, 6
8	**GemSpoolingDeactivated** Indicates that the entire spool has been transmitted or purged and messages are no longer being spooled.	6
9	**GemSpoolTransmitFailure** Indicates that a spooled message could not be transmitted.	6
10	**GemMessageAcknowledge** Indicates that a terminal message has acknowledged by operator.	N/A

Cooling Equipment Events

CEID	Description/Trigger	Default RPTID
101	**LoadPortReadyToLoad/ LoadPortEmpty** Indicates that load port is ready to load with carrier.	102
102	**LoadPortToBlocked** Indicates that load port have been loaded with carrier.	103
103	**LoadPortReadyToUnload** Indicates that carrier is ready for unload from load port.	104
104	**Job Status** Indicates the Job and Lot status.	101

APPENDIX D: DEFAULT REPORTS LIST

This section lists default RPTIDs, which the equipment has.

RPTID	Report Format	VID	VID Format	VID Type
1	1. Clock 2. ControlState 3. PreviousControlState	14 20 21	A U1 U1	SV SV SV
2	1. Clock 2. PPChangeName 3. PPChangeStatus	14 22 23	A A U1	SV DV DV
3	1. Clock 2. ProcessState 3. PreviousProcessState	14 43 42	A A A	SV SV SV
4	1. EventLimit 2. LimitVariable 3. TransitionType	27 28 29	B U4 B	DV DV DV
5	1. SpoolStartTime	37	A	SV
6	1. SpoolState 2. SpoolLoadSubstate 3. SpoolUnloadSubstate	38 36 39	U1 U1 U1	SV SV SV
101	1. Job ID and Lot ID Status	106	L	DV
102	1. Load Port ID	103	A	DV
103	1. Carrier ID and Load Port ID to be load	105	L	DV
104	1. Carrier ID and Load Port ID to be unload	107	L	DV

APPENDIX E: ALARM LIST

This section lists alarms, which the equipment has.

Error

ALID	ALTX
1	Didn't receive any StartJob from server after WaitingForStart send to server
2	Loading Time Expired !
3	Not Enough Available LoadPort
52	Staging Time Expired !
101	Unauthorized removal of Tray Carrier from the LoadPort (AA-01-01)
102	Unauthorized removal of Tray Carrier from the LoadPort (AA-01-02)
103	Unauthorized removal of Tray Carrier from the LoadPort (AA-01-03)
104	Unauthorized removal of Tray Carrier from the LoadPort (AA-01-04)
105	Unauthorized removal of Tray Carrier from the LoadPort (AA-01-05)
106	Unauthorized removal of Tray Carrier from the LoadPort (AA-02-01)
107	Unauthorized removal of Tray Carrier from the LoadPort (AA-02-02)
108	Unauthorized removal of Tray Carrier from the LoadPort (AA-02-03)
109	Unauthorized removal of Tray Carrier from the LoadPort (AA-02-04)
110	Unauthorized removal of Tray Carrier from the LoadPort (AA-02-05)
111	Unauthorized removal of Tray Carrier from the LoadPort (AA-03-01)
112	Unauthorized removal of Tray Carrier from the LoadPort (AA-03-02)
113	Unauthorized removal of Tray Carrier from the LoadPort (AA-03-03)
114	Unauthorized removal of Tray Carrier from the LoadPort (AA-03-04)
115	Unauthorized removal of Tray Carrier from the LoadPort (AA-03-05)
116	Unauthorized removal of Tray Carrier from the LoadPort (AA-04-01)
117	Unauthorized removal of Tray Carrier from the LoadPort (AA-04-02)
118	Unauthorized removal of Tray Carrier from the LoadPort (AA-04-03)
119	Unauthorized removal of Tray Carrier from the LoadPort (AA-04-04)
120	Unauthorized removal of Tray Carrier from the LoadPort (AA-04-05)
121	Unauthorized removal of Tray Carrier from the LoadPort (AA-05-01)
122	Unauthorized removal of Tray Carrier from the LoadPort (AA-05-02)
123	Unauthorized removal of Tray Carrier from the LoadPort (AA-05-03)
124	Unauthorized removal of Tray Carrier from the LoadPort (AA-05-04)
125	Unauthorized removal of Tray Carrier from the LoadPort (AA-05-05)
126	Unauthorized removal of Tray Carrier from the LoadPort (AA-06-01)
127	Unauthorized removal of Tray Carrier from the LoadPort (AA-06-02)
128	Unauthorized removal of Tray Carrier from the LoadPort (AA-06-03)
129	Unauthorized removal of Tray Carrier from the LoadPort (AA-06-04)
130	Unauthorized removal of Tray Carrier from the LoadPort (AA-06-05)

131	Unauthorized removal of Tray Carrier from the LoadPort (AA-07-01)
132	Unauthorized removal of Tray Carrier from the LoadPort (AA-07-02)
133	Unauthorized removal of Tray Carrier from the LoadPort (AA-07-03)
134	Unauthorized removal of Tray Carrier from the LoadPort (AA-07-04)
135	Unauthorized removal of Tray Carrier from the LoadPort (AA-07-05)
136	Unauthorized removal of Tray Carrier from the LoadPort (AB-01-01)
137	Unauthorized removal of Tray Carrier from the LoadPort (AB-01-02)
138	Unauthorized removal of Tray Carrier from the LoadPort (AB-01-03)
139	Unauthorized removal of Tray Carrier from the LoadPort (AB-01-04)
140	Unauthorized removal of Tray Carrier from the LoadPort (AB-01-05)
141	Unauthorized removal of Tray Carrier from the LoadPort (AB-02-01)
142	Unauthorized removal of Tray Carrier from the LoadPort (AB-02-02)
143	Unauthorized removal of Tray Carrier from the LoadPort (AB-02-03)
144	Unauthorized removal of Tray Carrier from the LoadPort (AB-02-04)
145	Unauthorized removal of Tray Carrier from the LoadPort (AB-02-05)
146	Unauthorized removal of Tray Carrier from the LoadPort (AB-03-01)
147	Unauthorized removal of Tray Carrier from the LoadPort (AB-03-02)
148	Unauthorized removal of Tray Carrier from the LoadPort (AB-03-03)
149	Unauthorized removal of Tray Carrier from the LoadPort (AB-03-04)
150	Unauthorized removal of Tray Carrier from the LoadPort (AB-03-05)
151	Unauthorized removal of Tray Carrier from the LoadPort (AB-04-01)
152	Unauthorized removal of Tray Carrier from the LoadPort (AB-04-02)
153	Unauthorized removal of Tray Carrier from the LoadPort (AB-04-03)
154	Unauthorized removal of Tray Carrier from the LoadPort (AB-04-04)
155	Unauthorized removal of Tray Carrier from the LoadPort (AB-04-05)
156	Unauthorized removal of Tray Carrier from the LoadPort (AB-05-01)
157	Unauthorized removal of Tray Carrier from the LoadPort (AB-05-02)
158	Unauthorized removal of Tray Carrier from the LoadPort (AB-05-03)
159	Unauthorized removal of Tray Carrier from the LoadPort (AB-05-04)
160	Unauthorized removal of Tray Carrier from the LoadPort (AB-05-05)
161	Unauthorized removal of Tray Carrier from the LoadPort (AB-06-01)
162	Unauthorized removal of Tray Carrier from the LoadPort (AB-06-02)
163	Unauthorized removal of Tray Carrier from the LoadPort (AB-06-03)
164	Unauthorized removal of Tray Carrier from the LoadPort (AB-06-04)
165	Unauthorized removal of Tray Carrier from the LoadPort (AB-06-05)
166	Unauthorized removal of Tray Carrier from the LoadPort (AB-07-01)
167	Unauthorized removal of Tray Carrier from the LoadPort (AB-07-02)
168	Unauthorized removal of Tray Carrier from the LoadPort (AB-07-03)
169	Unauthorized removal of Tray Carrier from the LoadPort (AB-07-04)
170	Unauthorized removal of Tray Carrier from the LoadPort (AB-07-05)

171	Unauthorized removal of Tray Carrier from the LoadPort (AC-01-01)
172	Unauthorized removal of Tray Carrier from the LoadPort (AC-01-02)
173	Unauthorized removal of Tray Carrier from the LoadPort (AC-01-03)
174	Unauthorized removal of Tray Carrier from the LoadPort (AC-01-04)
175	Unauthorized removal of Tray Carrier from the LoadPort (AC-01-05)
176	Unauthorized removal of Tray Carrier from the LoadPort (AC-02-01)
177	Unauthorized removal of Tray Carrier from the LoadPort (AC-02-02)
178	Unauthorized removal of Tray Carrier from the LoadPort (AC-02-03)
179	Unauthorized removal of Tray Carrier from the LoadPort (AC-02-04)
180	Unauthorized removal of Tray Carrier from the LoadPort (AC-02-05)
181	Unauthorized removal of Tray Carrier from the LoadPort (AC-03-01)
182	Unauthorized removal of Tray Carrier from the LoadPort (AC-03-02)
183	Unauthorized removal of Tray Carrier from the LoadPort (AC-03-03)
184	Unauthorized removal of Tray Carrier from the LoadPort (AC-03-04)
185	Unauthorized removal of Tray Carrier from the LoadPort (AC-03-05)
186	Unauthorized removal of Tray Carrier from the LoadPort (AC-04-01)
187	Unauthorized removal of Tray Carrier from the LoadPort (AC-04-02)
188	Unauthorized removal of Tray Carrier from the LoadPort (AC-04-03)
189	Unauthorized removal of Tray Carrier from the LoadPort (AC-04-04)
190	Unauthorized removal of Tray Carrier from the LoadPort (AC-04-05)
191	Unauthorized removal of Tray Carrier from the LoadPort (AC-05-01)
192	Unauthorized removal of Tray Carrier from the LoadPort (AC-05-02)
193	Unauthorized removal of Tray Carrier from the LoadPort (AC-05-03)
194	Unauthorized removal of Tray Carrier from the LoadPort (AC-05-04)
195	Unauthorized removal of Tray Carrier from the LoadPort (AC-05-05)
196	Unauthorized removal of Tray Carrier from the LoadPort (AC-06-01)
197	Unauthorized removal of Tray Carrier from the LoadPort (AC-06-02)
198	Unauthorized removal of Tray Carrier from the LoadPort (AC-06-03)
199	Unauthorized removal of Tray Carrier from the LoadPort (AC-06-04)
200	Unauthorized removal of Tray Carrier from the LoadPort (AC-06-05)
201	Unauthorized removal of Tray Carrier from the LoadPort (AC-07-01)
202	Unauthorized removal of Tray Carrier from the LoadPort (AC-07-02)
203	Unauthorized removal of Tray Carrier from the LoadPort (AC-07-03)
204	Unauthorized removal of Tray Carrier from the LoadPort (AC-07-04)
205	Unauthorized removal of Tray Carrier from the LoadPort (AC-07-05)
301	Unauthorized Tray Carrier inside the LoadPort (AA-01-01)
302	Unauthorized Tray Carrier inside the LoadPort (AA-01-02)
303	Unauthorized Tray Carrier inside the LoadPort (AA-01-03)
304	Unauthorized Tray Carrier inside the LoadPort (AA-01-04)
305	Unauthorized Tray Carrier inside the LoadPort (AA-01-05)

306	Unauthorized Tray Carrier inside the LoadPort (AA-02-01)
307	Unauthorized Tray Carrier inside the LoadPort (AA-02-02)
308	Unauthorized Tray Carrier inside the LoadPort (AA-02-03)
309	Unauthorized Tray Carrier inside the LoadPort (AA-02-04)
310	Unauthorized Tray Carrier inside the LoadPort (AA-02-05)
311	Unauthorized Tray Carrier inside the LoadPort (AA-03-01)
312	Unauthorized Tray Carrier inside the LoadPort (AA-03-02)
313	Unauthorized Tray Carrier inside the LoadPort (AA-03-03)
314	Unauthorized Tray Carrier inside the LoadPort (AA-03-04)
315	Unauthorized Tray Carrier inside the LoadPort (AA-03-05)
316	Unauthorized Tray Carrier inside the LoadPort (AA-04-01)
317	Unauthorized Tray Carrier inside the LoadPort (AA-04-02)
318	Unauthorized Tray Carrier inside the LoadPort (AA-04-03)
319	Unauthorized Tray Carrier inside the LoadPort (AA-04-04)
320	Unauthorized Tray Carrier inside the LoadPort (AA-04-05)
321	Unauthorized Tray Carrier inside the LoadPort (AA-05-01)
322	Unauthorized Tray Carrier inside the LoadPort (AA-05-02)
323	Unauthorized Tray Carrier inside the LoadPort (AA-05-03)
324	Unauthorized Tray Carrier inside the LoadPort (AA-05-04)
325	Unauthorized Tray Carrier inside the LoadPort (AA-05-05)
326	Unauthorized Tray Carrier inside the LoadPort (AA-06-01)
327	Unauthorized Tray Carrier inside the LoadPort (AA-06-02)
328	Unauthorized Tray Carrier inside the LoadPort (AA-06-03)
329	Unauthorized Tray Carrier inside the LoadPort (AA-06-04)
330	Unauthorized Tray Carrier inside the LoadPort (AA-06-05)
331	Unauthorized Tray Carrier inside the LoadPort (AA-07-01)
332	Unauthorized Tray Carrier inside the LoadPort (AA-07-02)
333	Unauthorized Tray Carrier inside the LoadPort (AA-07-03)
334	Unauthorized Tray Carrier inside the LoadPort (AA-07-04)
335	Unauthorized Tray Carrier inside the LoadPort (AA-07-05)
336	Unauthorized Tray Carrier inside the LoadPort (AB-01-01)
337	Unauthorized Tray Carrier inside the LoadPort (AB-01-02)
338	Unauthorized Tray Carrier inside the LoadPort (AB-01-03)
339	Unauthorized Tray Carrier inside the LoadPort (AB-01-04)
340	Unauthorized Tray Carrier inside the LoadPort (AB-01-05)
341	Unauthorized Tray Carrier inside the LoadPort (AB-02-01)
342	Unauthorized Tray Carrier inside the LoadPort (AB-02-02)
343	Unauthorized Tray Carrier inside the LoadPort (AB-02-03)
344	Unauthorized Tray Carrier inside the LoadPort (AB-02-04)
345	Unauthorized Tray Carrier inside the LoadPort (AB-02-05)

346	Unauthorized Tray Carrier inside the LoadPort (AB-03-01)
347	Unauthorized Tray Carrier inside the LoadPort (AB-03-02)
348	Unauthorized Tray Carrier inside the LoadPort (AB-03-03)
349	Unauthorized Tray Carrier inside the LoadPort (AB-03-04)
350	Unauthorized Tray Carrier inside the LoadPort (AB-03-05)
351	Unauthorized Tray Carrier inside the LoadPort (AB-04-01)
352	Unauthorized Tray Carrier inside the LoadPort (AB-04-02)
353	Unauthorized Tray Carrier inside the LoadPort (AB-04-03)
354	Unauthorized Tray Carrier inside the LoadPort (AB-04-04)
355	Unauthorized Tray Carrier inside the LoadPort (AB-04-05)
356	Unauthorized Tray Carrier inside the LoadPort (AB-05-01)
357	Unauthorized Tray Carrier inside the LoadPort (AB-05-02)
358	Unauthorized Tray Carrier inside the LoadPort (AB-05-03)
359	Unauthorized Tray Carrier inside the LoadPort (AB-05-04)
360	Unauthorized Tray Carrier inside the LoadPort (AB-05-05)
361	Unauthorized Tray Carrier inside the LoadPort (AB-06-01)
362	Unauthorized Tray Carrier inside the LoadPort (AB-06-02)
363	Unauthorized Tray Carrier inside the LoadPort (AB-06-03)
364	Unauthorized Tray Carrier inside the LoadPort (AB-06-04)
365	Unauthorized Tray Carrier inside the LoadPort (AB-06-05)
366	Unauthorized Tray Carrier inside the LoadPort (AB-07-01)
367	Unauthorized Tray Carrier inside the LoadPort (AB-07-02)
368	Unauthorized Tray Carrier inside the LoadPort (AB-07-03)
369	Unauthorized Tray Carrier inside the LoadPort (AB-07-04)
370	Unauthorized Tray Carrier inside the LoadPort (AB-07-05)
371	Unauthorized Tray Carrier inside the LoadPort (AC-01-01)
372	Unauthorized Tray Carrier inside the LoadPort (AC-01-02)
373	Unauthorized Tray Carrier inside the LoadPort (AC-01-03)
374	Unauthorized Tray Carrier inside the LoadPort (AC-01-04)
375	Unauthorized Tray Carrier inside the LoadPort (AC-01-05)
376	Unauthorized Tray Carrier inside the LoadPort (AC-02-01)
377	Unauthorized Tray Carrier inside the LoadPort (AC-02-02)
378	Unauthorized Tray Carrier inside the LoadPort (AC-02-03)
379	Unauthorized Tray Carrier inside the LoadPort (AC-02-04)
380	Unauthorized Tray Carrier inside the LoadPort (AC-02-05)
381	Unauthorized Tray Carrier inside the LoadPort (AC-03-01)
382	Unauthorized Tray Carrier inside the LoadPort (AC-03-02)
383	Unauthorized Tray Carrier inside the LoadPort (AC-03-03)
384	Unauthorized Tray Carrier inside the LoadPort (AC-03-04)
385	Unauthorized Tray Carrier inside the LoadPort (AC-03-05)

386	Unauthorized Tray Carrier inside the LoadPort (AC-04-01)
387	Unauthorized Tray Carrier inside the LoadPort (AC-04-02)
388	Unauthorized Tray Carrier inside the LoadPort (AC-04-03)
389	Unauthorized Tray Carrier inside the LoadPort (AC-04-04)
390	Unauthorized Tray Carrier inside the LoadPort (AC-04-05)
391	Unauthorized Tray Carrier inside the LoadPort (AC-05-01)
392	Unauthorized Tray Carrier inside the LoadPort (AC-05-02)
393	Unauthorized Tray Carrier inside the LoadPort (AC-05-03)
394	Unauthorized Tray Carrier inside the LoadPort (AC-05-04)
395	Unauthorized Tray Carrier inside the LoadPort (AC-05-05)
396	Unauthorized Tray Carrier inside the LoadPort (AC-06-01)
397	Unauthorized Tray Carrier inside the LoadPort (AC-06-02)
398	Unauthorized Tray Carrier inside the LoadPort (AC-06-03)
399	Unauthorized Tray Carrier inside the LoadPort (AC-06-04)
400	Unauthorized Tray Carrier inside the LoadPort (AC-06-05)
401	Unauthorized Tray Carrier inside the LoadPort (AC-07-01)
402	Unauthorized Tray Carrier inside the LoadPort (AC-07-02)
403	Unauthorized Tray Carrier inside the LoadPort (AC-07-03)
404	Unauthorized Tray Carrier inside the LoadPort (AC-07-04)
405	Unauthorized Tray Carrier inside the LoadPort (AC-07-05)
501	Unauthorized removal of Tube Carrier from the LoadPort (BA-01-01)
502	Unauthorized removal of Tube Carrier from the LoadPort (BA-01-02)
503	Unauthorized removal of Tube Carrier from the LoadPort (BA-01-03)
504	Unauthorized removal of Tube Carrier from the LoadPort (BA-01-04)
505	Unauthorized removal of Tube Carrier from the LoadPort (BA-01-05)
506	Unauthorized removal of Tube Carrier from the LoadPort (BA-01-06)
507	Unauthorized removal of Tube Carrier from the LoadPort (BA-02-01)
508	Unauthorized removal of Tube Carrier from the LoadPort (BA-02-02)
509	Unauthorized removal of Tube Carrier from the LoadPort (BA-02-03)
510	Unauthorized removal of Tube Carrier from the LoadPort (BA-02-04)
511	Unauthorized removal of Tube Carrier from the LoadPort (BA-02-05)
512	Unauthorized removal of Tube Carrier from the LoadPort (BA-02-06)
513	Unauthorized removal of Tube Carrier from the LoadPort (BA-03-01)
514	Unauthorized removal of Tube Carrier from the LoadPort (BA-03-02)
515	Unauthorized removal of Tube Carrier from the LoadPort (BA-03-03)
516	Unauthorized removal of Tube Carrier from the LoadPort (BA-03-04)
517	Unauthorized removal of Tube Carrier from the LoadPort (BA-03-05)
518	Unauthorized removal of Tube Carrier from the LoadPort (BA-03-06)
519	Unauthorized removal of Tube Carrier from the LoadPort (BA-04-01)
520	Unauthorized removal of Tube Carrier from the LoadPort (BA-04-02)

521	Unauthorized removal of Tube Carrier from the LoadPort (BA-04-03)
522	Unauthorized removal of Tube Carrier from the LoadPort (BA-04-04)
523	Unauthorized removal of Tube Carrier from the LoadPort (BA-04-05)
524	Unauthorized removal of Tube Carrier from the LoadPort (BA-04-06)
525	Unauthorized removal of Tube Carrier from the LoadPort (BA-05-01)
526	Unauthorized removal of Tube Carrier from the LoadPort (BA-05-02)
527	Unauthorized removal of Tube Carrier from the LoadPort (BA-05-03)
528	Unauthorized removal of Tube Carrier from the LoadPort (BA-05-04)
529	Unauthorized removal of Tube Carrier from the LoadPort (BA-05-05)
530	Unauthorized removal of Tube Carrier from the LoadPort (BA-05-06)
531	Unauthorized removal of Tube Carrier from the LoadPort (BA-06-01)
532	Unauthorized removal of Tube Carrier from the LoadPort (BA-06-02)
533	Unauthorized removal of Tube Carrier from the LoadPort (BA-06-03)
534	Unauthorized removal of Tube Carrier from the LoadPort (BA-06-04)
535	Unauthorized removal of Tube Carrier from the LoadPort (BA-06-05)
536	Unauthorized removal of Tube Carrier from the LoadPort (BA-06-06)
537	Unauthorized removal of Tube Carrier from the LoadPort (BA-07-01)
538	Unauthorized removal of Tube Carrier from the LoadPort (BA-07-02)
539	Unauthorized removal of Tube Carrier from the LoadPort (BA-07-03)
540	Unauthorized removal of Tube Carrier from the LoadPort (BA-07-04)
541	Unauthorized removal of Tube Carrier from the LoadPort (BA-07-05)
542	Unauthorized removal of Tube Carrier from the LoadPort (BA-07-06)
543	Unauthorized removal of Tube Carrier from the LoadPort (BA-08-01)
544	Unauthorized removal of Tube Carrier from the LoadPort (BA-08-02)
545	Unauthorized removal of Tube Carrier from the LoadPort (BA-08-03)
546	Unauthorized removal of Tube Carrier from the LoadPort (BA-08-04)
547	Unauthorized removal of Tube Carrier from the LoadPort (BA-08-05)
548	Unauthorized removal of Tube Carrier from the LoadPort (BA-08-06)
549	Unauthorized removal of Tube Carrier from the LoadPort (BB-01-01)
550	Unauthorized removal of Tube Carrier from the LoadPort (BB-01-02)
551	Unauthorized removal of Tube Carrier from the LoadPort (BB-01-03)
552	Unauthorized removal of Tube Carrier from the LoadPort (BB-01-04)
553	Unauthorized removal of Tube Carrier from the LoadPort (BB-01-05)
554	Unauthorized removal of Tube Carrier from the LoadPort (BB-01-06)
555	Unauthorized removal of Tube Carrier from the LoadPort (BB-02-01)
556	Unauthorized removal of Tube Carrier from the LoadPort (BB-02-02)
557	Unauthorized removal of Tube Carrier from the LoadPort (BB-02-03)
558	Unauthorized removal of Tube Carrier from the LoadPort (BB-02-04)
559	Unauthorized removal of Tube Carrier from the LoadPort (BB-02-05)
560	Unauthorized removal of Tube Carrier from the LoadPort (BB-02-06)

561	Unauthorized removal of Tube Carrier from the LoadPort (BB-03-01)
562	Unauthorized removal of Tube Carrier from the LoadPort (BB-03-02)
563	Unauthorized removal of Tube Carrier from the LoadPort (BB-03-03)
564	Unauthorized removal of Tube Carrier from the LoadPort (BB-03-04)
565	Unauthorized removal of Tube Carrier from the LoadPort (BB-03-05)
566	Unauthorized removal of Tube Carrier from the LoadPort (BB-03-06)
567	Unauthorized removal of Tube Carrier from the LoadPort (BB-04-01)
568	Unauthorized removal of Tube Carrier from the LoadPort (BB-04-02)
569	Unauthorized removal of Tube Carrier from the LoadPort (BB-04-03)
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572	Unauthorized removal of Tube Carrier from the LoadPort (BB-04-06)
573	Unauthorized removal of Tube Carrier from the LoadPort (BB-05-01)
574	Unauthorized removal of Tube Carrier from the LoadPort (BB-05-02)
575	Unauthorized removal of Tube Carrier from the LoadPort (BB-05-03)
576	Unauthorized removal of Tube Carrier from the LoadPort (BB-05-04)
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578	Unauthorized removal of Tube Carrier from the LoadPort (BB-05-06)
579	Unauthorized removal of Tube Carrier from the LoadPort (BB-06-01)
580	Unauthorized removal of Tube Carrier from the LoadPort (BB-06-02)
581	Unauthorized removal of Tube Carrier from the LoadPort (BB-06-03)
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586	Unauthorized removal of Tube Carrier from the LoadPort (BB-07-02)
587	Unauthorized removal of Tube Carrier from the LoadPort (BB-07-03)
588	Unauthorized removal of Tube Carrier from the LoadPort (BB-07-04)
589	Unauthorized removal of Tube Carrier from the LoadPort (BB-07-05)
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591	Unauthorized removal of Tube Carrier from the LoadPort (BB-08-01)
592	Unauthorized removal of Tube Carrier from the LoadPort (BB-08-02)
593	Unauthorized removal of Tube Carrier from the LoadPort (BB-08-03)
594	Unauthorized removal of Tube Carrier from the LoadPort (BB-08-04)
595	Unauthorized removal of Tube Carrier from the LoadPort (BB-08-05)
596	Unauthorized removal of Tube Carrier from the LoadPort (BB-08-06)
601	Unauthorized Tube Carrier inside the LoadPort (BA-01-01)
602	Unauthorized Tube Carrier inside the LoadPort (BA-01-02)
603	Unauthorized Tube Carrier inside the LoadPort (BA-01-03)
604	Unauthorized Tube Carrier inside the LoadPort (BA-01-04)

605	Unauthorized Tube Carrier inside the LoadPort (BA-01-05)
606	Unauthorized Tube Carrier inside the LoadPort (BA-01-06)
607	Unauthorized Tube Carrier inside the LoadPort (BA-02-01)
608	Unauthorized Tube Carrier inside the LoadPort (BA-02-02)
609	Unauthorized Tube Carrier inside the LoadPort (BA-02-03)
610	Unauthorized Tube Carrier inside the LoadPort (BA-02-04)
611	Unauthorized Tube Carrier inside the LoadPort (BA-02-05)
612	Unauthorized Tube Carrier inside the LoadPort (BA-02-06)
613	Unauthorized Tube Carrier inside the LoadPort (BA-03-01)
614	Unauthorized Tube Carrier inside the LoadPort (BA-03-02)
615	Unauthorized Tube Carrier inside the LoadPort (BA-03-03)
616	Unauthorized Tube Carrier inside the LoadPort (BA-03-04)
617	Unauthorized Tube Carrier inside the LoadPort (BA-03-05)
618	Unauthorized Tube Carrier inside the LoadPort (BA-03-06)
619	Unauthorized Tube Carrier inside the LoadPort (BA-04-01)
620	Unauthorized Tube Carrier inside the LoadPort (BA-04-02)
621	Unauthorized Tube Carrier inside the LoadPort (BA-04-03)
622	Unauthorized Tube Carrier inside the LoadPort (BA-04-04)
623	Unauthorized Tube Carrier inside the LoadPort (BA-04-05)
624	Unauthorized Tube Carrier inside the LoadPort (BA-04-06)
625	Unauthorized Tube Carrier inside the LoadPort (BA-05-01)
626	Unauthorized Tube Carrier inside the LoadPort (BA-05-02)
627	Unauthorized Tube Carrier inside the LoadPort (BA-05-03)
628	Unauthorized Tube Carrier inside the LoadPort (BA-05-04)
629	Unauthorized Tube Carrier inside the LoadPort (BA-05-05)
630	Unauthorized Tube Carrier inside the LoadPort (BA-05-06)
631	Unauthorized Tube Carrier inside the LoadPort (BA-06-01)
632	Unauthorized Tube Carrier inside the LoadPort (BA-06-02)
633	Unauthorized Tube Carrier inside the LoadPort (BA-06-03)
634	Unauthorized Tube Carrier inside the LoadPort (BA-06-04)
635	Unauthorized Tube Carrier inside the LoadPort (BA-06-05)
636	Unauthorized Tube Carrier inside the LoadPort (BA-06-06)
637	Unauthorized Tube Carrier inside the LoadPort (BA-07-01)
638	Unauthorized Tube Carrier inside the LoadPort (BA-07-02)
639	Unauthorized Tube Carrier inside the LoadPort (BA-07-03)
640	Unauthorized Tube Carrier inside the LoadPort (BA-07-04)
641	Unauthorized Tube Carrier inside the LoadPort (BA-07-05)
642	Unauthorized Tube Carrier inside the LoadPort (BA-07-06)
643	Unauthorized Tube Carrier inside the LoadPort (BA-08-01)
644	Unauthorized Tube Carrier inside the LoadPort (BA-08-02)

645	Unauthorized Tube Carrier inside the LoadPort (BA-08-03)
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649	Unauthorized Tube Carrier inside the LoadPort (BB-01-01)
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652	Unauthorized Tube Carrier inside the LoadPort (BB-01-04)
653	Unauthorized Tube Carrier inside the LoadPort (BB-01-05)
654	Unauthorized Tube Carrier inside the LoadPort (BB-01-06)
655	Unauthorized Tube Carrier inside the LoadPort (BB-02-01)
656	Unauthorized Tube Carrier inside the LoadPort (BB-02-02)
657	Unauthorized Tube Carrier inside the LoadPort (BB-02-03)
658	Unauthorized Tube Carrier inside the LoadPort (BB-02-04)
659	Unauthorized Tube Carrier inside the LoadPort (BB-02-05)
660	Unauthorized Tube Carrier inside the LoadPort (BB-02-06)
661	Unauthorized Tube Carrier inside the LoadPort (BB-03-01)
662	Unauthorized Tube Carrier inside the LoadPort (BB-03-02)
663	Unauthorized Tube Carrier inside the LoadPort (BB-03-03)
664	Unauthorized Tube Carrier inside the LoadPort (BB-03-04)
665	Unauthorized Tube Carrier inside the LoadPort (BB-03-05)
666	Unauthorized Tube Carrier inside the LoadPort (BB-03-06)
667	Unauthorized Tube Carrier inside the LoadPort (BB-04-01)
668	Unauthorized Tube Carrier inside the LoadPort (BB-04-02)
669	Unauthorized Tube Carrier inside the LoadPort (BB-04-03)
670	Unauthorized Tube Carrier inside the LoadPort (BB-04-04)
671	Unauthorized Tube Carrier inside the LoadPort (BB-04-05)
672	Unauthorized Tube Carrier inside the LoadPort (BB-04-06)
673	Unauthorized Tube Carrier inside the LoadPort (BB-05-01)
674	Unauthorized Tube Carrier inside the LoadPort (BB-05-02)
675	Unauthorized Tube Carrier inside the LoadPort (BB-05-03)
676	Unauthorized Tube Carrier inside the LoadPort (BB-05-04)
677	Unauthorized Tube Carrier inside the LoadPort (BB-05-05)
678	Unauthorized Tube Carrier inside the LoadPort (BB-05-06)
679	Unauthorized Tube Carrier inside the LoadPort (BB-06-01)
680	Unauthorized Tube Carrier inside the LoadPort (BB-06-02)
681	Unauthorized Tube Carrier inside the LoadPort (BB-06-03)
682	Unauthorized Tube Carrier inside the LoadPort (BB-06-04)
683	Unauthorized Tube Carrier inside the LoadPort (BB-06-05)
684	Unauthorized Tube Carrier inside the LoadPort (BB-06-06)

685	Unauthorized Tube Carrier inside the LoadPort (BB-07-01)
686	Unauthorized Tube Carrier inside the LoadPort (BB-07-02)
687	Unauthorized Tube Carrier inside the LoadPort (BB-07-03)
688	Unauthorized Tube Carrier inside the LoadPort (BB-07-04)
689	Unauthorized Tube Carrier inside the LoadPort (BB-07-05)
690	Unauthorized Tube Carrier inside the LoadPort (BB-07-06)
691	Unauthorized Tube Carrier inside the LoadPort (BB-08-01)
692	Unauthorized Tube Carrier inside the LoadPort (BB-08-02)
693	Unauthorized Tube Carrier inside the LoadPort (BB-08-03)
694	Unauthorized Tube Carrier inside the LoadPort (BB-08-04)
695	Unauthorized Tube Carrier inside the LoadPort (BB-08-05)
696	Unauthorized Tube Carrier inside the LoadPort (BB-08-06)

Warning

ALID	ALTX
1001	Carrier ID scanned not assign to any Job
1002	Loading Process have been blocked due to pending Unloading Process to complete
1003	Loading Process pending complete
1004	StartJob Command Received but wrong Job ID
1005	Loading Process have been blocked due to error occurs
1051	Lot No scanned not assign to any Job
1052	Unloading Process have been blocked due to pending Loading Process to complete
1053	Unloading Process pending complete
1054	Job is not ready for Unloading
1055	Unloading Process have been blocked due to error occurs
1071	PC can't communicate with Controllers (10.0.50.110)
1072	PC can't communicate with Controllers (10.0.50.120)
1101	PC can't communicate with LoadPort (AA-01-01)
1102	PC can't communicate with LoadPort (AA-01-02)
1103	PC can't communicate with LoadPort (AA-01-03)
1104	PC can't communicate with LoadPort (AA-01-04)
1105	PC can't communicate with LoadPort (AA-01-05)
1106	PC can't communicate with LoadPort (AA-02-01)
1107	PC can't communicate with LoadPort (AA-02-02)
1108	PC can't communicate with LoadPort (AA-02-03)
1109	PC can't communicate with LoadPort (AA-02-04)
1110	PC can't communicate with LoadPort (AA-02-05)
1111	PC can't communicate with LoadPort (AA-03-01)
1112	PC can't communicate with LoadPort (AA-03-02)
1113	PC can't communicate with LoadPort (AA-03-03)
1114	PC can't communicate with LoadPort (AA-03-04)
1115	PC can't communicate with LoadPort (AA-03-05)
1116	PC can't communicate with LoadPort (AA-04-01)
1117	PC can't communicate with LoadPort (AA-04-02)
1118	PC can't communicate with LoadPort (AA-04-03)
1119	PC can't communicate with LoadPort (AA-04-04)
1120	PC can't communicate with LoadPort (AA-04-05)
1121	PC can't communicate with LoadPort (AA-05-01)
1122	PC can't communicate with LoadPort (AA-05-02)
1123	PC can't communicate with LoadPort (AA-05-03)
1124	PC can't communicate with LoadPort (AA-05-04)

1125	PC can't communicate with LoadPort (AA-05-05)
1126	PC can't communicate with LoadPort (AA-06-01)
1127	PC can't communicate with LoadPort (AA-06-02)
1128	PC can't communicate with LoadPort (AA-06-03)
1129	PC can't communicate with LoadPort (AA-06-04)
1130	PC can't communicate with LoadPort (AA-06-05)
1131	PC can't communicate with LoadPort (AA-07-01)
1132	PC can't communicate with LoadPort (AA-07-02)
1133	PC can't communicate with LoadPort (AA-07-03)
1134	PC can't communicate with LoadPort (AA-07-04)
1135	PC can't communicate with LoadPort (AA-07-05)
1136	PC can't communicate with LoadPort (AB-01-01)
1137	PC can't communicate with LoadPort (AB-01-02)
1138	PC can't communicate with LoadPort (AB-01-03)
1139	PC can't communicate with LoadPort (AB-01-04)
1140	PC can't communicate with LoadPort (AB-01-05)
1141	PC can't communicate with LoadPort (AB-02-01)
1142	PC can't communicate with LoadPort (AB-02-02)
1143	PC can't communicate with LoadPort (AB-02-03)
1144	PC can't communicate with LoadPort (AB-02-04)
1145	PC can't communicate with LoadPort (AB-02-05)
1146	PC can't communicate with LoadPort (AB-03-01)
1147	PC can't communicate with LoadPort (AB-03-02)
1148	PC can't communicate with LoadPort (AB-03-03)
1149	PC can't communicate with LoadPort (AB-03-04)
1150	PC can't communicate with LoadPort (AB-03-05)
1151	PC can't communicate with LoadPort (AB-04-01)
1152	PC can't communicate with LoadPort (AB-04-02)
1153	PC can't communicate with LoadPort (AB-04-03)
1154	PC can't communicate with LoadPort (AB-04-04)
1155	PC can't communicate with LoadPort (AB-04-05)
1156	PC can't communicate with LoadPort (AB-05-01)
1157	PC can't communicate with LoadPort (AB-05-02)
1158	PC can't communicate with LoadPort (AB-05-03)
1159	PC can't communicate with LoadPort (AB-05-04)
1160	PC can't communicate with LoadPort (AB-05-05)
1161	PC can't communicate with LoadPort (AB-06-01)
1162	PC can't communicate with LoadPort (AB-06-02)
1163	PC can't communicate with LoadPort (AB-06-03)
1164	PC can't communicate with LoadPort (AB-06-04)

1165	PC can't communicate with LoadPort (AB-06-05)
1166	PC can't communicate with LoadPort (AB-07-01)
1167	PC can't communicate with LoadPort (AB-07-02)
1168	PC can't communicate with LoadPort (AB-07-03)
1169	PC can't communicate with LoadPort (AB-07-04)
1170	PC can't communicate with LoadPort (AB-07-05)
1171	PC can't communicate with LoadPort (AC-01-01)
1172	PC can't communicate with LoadPort (AC-01-02)
1173	PC can't communicate with LoadPort (AC-01-03)
1174	PC can't communicate with LoadPort (AC-01-04)
1175	PC can't communicate with LoadPort (AC-01-05)
1176	PC can't communicate with LoadPort (AC-02-01)
1177	PC can't communicate with LoadPort (AC-02-02)
1178	PC can't communicate with LoadPort (AC-02-03)
1179	PC can't communicate with LoadPort (AC-02-04)
1180	PC can't communicate with LoadPort (AC-02-05)
1181	PC can't communicate with LoadPort (AC-03-01)
1182	PC can't communicate with LoadPort (AC-03-02)
1183	PC can't communicate with LoadPort (AC-03-03)
1184	PC can't communicate with LoadPort (AC-03-04)
1185	PC can't communicate with LoadPort (AC-03-05)
1186	PC can't communicate with LoadPort (AC-04-01)
1187	PC can't communicate with LoadPort (AC-04-02)
1188	PC can't communicate with LoadPort (AC-04-03)
1189	PC can't communicate with LoadPort (AC-04-04)
1190	PC can't communicate with LoadPort (AC-04-05)
1191	PC can't communicate with LoadPort (AC-05-01)
1192	PC can't communicate with LoadPort (AC-05-02)
1193	PC can't communicate with LoadPort (AC-05-03)
1194	PC can't communicate with LoadPort (AC-05-04)
1195	PC can't communicate with LoadPort (AC-05-05)
1196	PC can't communicate with LoadPort (AC-06-01)
1197	PC can't communicate with LoadPort (AC-06-02)
1198	PC can't communicate with LoadPort (AC-06-03)
1199	PC can't communicate with LoadPort (AC-06-04)
1200	PC can't communicate with LoadPort (AC-06-05)
1201	PC can't communicate with LoadPort (AC-07-01)
1202	PC can't communicate with LoadPort (AC-07-02)
1203	PC can't communicate with LoadPort (AC-07-03)
1204	PC can't communicate with LoadPort (AC-07-04)

1205	PC can't communicate with LoadPort (AC-07-05)
1501	PC can't communicate with LoadPort (BA-01-01)
1502	PC can't communicate with LoadPort (BA-01-02)
1503	PC can't communicate with LoadPort (BA-01-03)
1504	PC can't communicate with LoadPort (BA-01-04)
1505	PC can't communicate with LoadPort (BA-01-05)
1506	PC can't communicate with LoadPort (BA-01-06)
1507	PC can't communicate with LoadPort (BA-02-01)
1508	PC can't communicate with LoadPort (BA-02-02)
1509	PC can't communicate with LoadPort (BA-02-03)
1510	PC can't communicate with LoadPort (BA-02-04)
1511	PC can't communicate with LoadPort (BA-02-05)
1512	PC can't communicate with LoadPort (BA-02-06)
1513	PC can't communicate with LoadPort (BA-03-01)
1514	PC can't communicate with LoadPort (BA-03-02)
1515	PC can't communicate with LoadPort (BA-03-03)
1516	PC can't communicate with LoadPort (BA-03-04)
1517	PC can't communicate with LoadPort (BA-03-05)
1518	PC can't communicate with LoadPort (BA-03-06)
1519	PC can't communicate with LoadPort (BA-04-01)
1520	PC can't communicate with LoadPort (BA-04-02)
1521	PC can't communicate with LoadPort (BA-04-03)
1522	PC can't communicate with LoadPort (BA-04-04)
1523	PC can't communicate with LoadPort (BA-04-05)
1524	PC can't communicate with LoadPort (BA-04-06)
1525	PC can't communicate with LoadPort (BA-05-01)
1526	PC can't communicate with LoadPort (BA-05-02)
1527	PC can't communicate with LoadPort (BA-05-03)
1528	PC can't communicate with LoadPort (BA-05-04)
1529	PC can't communicate with LoadPort (BA-05-05)
1530	PC can't communicate with LoadPort (BA-05-06)
1531	PC can't communicate with LoadPort (BA-06-01)
1532	PC can't communicate with LoadPort (BA-06-02)
1533	PC can't communicate with LoadPort (BA-06-03)
1534	PC can't communicate with LoadPort (BA-06-04)
1535	PC can't communicate with LoadPort (BA-06-05)
1536	PC can't communicate with LoadPort (BA-06-06)
1537	PC can't communicate with LoadPort (BA-07-01)
1538	PC can't communicate with LoadPort (BA-07-02)
1539	PC can't communicate with LoadPort (BA-07-03)

1540	PC can't communicate with LoadPort (BA-07-04)
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1542	PC can't communicate with LoadPort (BA-07-06)
1543	PC can't communicate with LoadPort (BA-08-01)
1544	PC can't communicate with LoadPort (BA-08-02)
1545	PC can't communicate with LoadPort (BA-08-03)
1546	PC can't communicate with LoadPort (BA-08-04)
1547	PC can't communicate with LoadPort (BA-08-05)
1548	PC can't communicate with LoadPort (BA-08-06)
1549	PC can't communicate with LoadPort (BB-01-01)
1550	PC can't communicate with LoadPort (BB-01-02)
1551	PC can't communicate with LoadPort (BB-01-03)
1552	PC can't communicate with LoadPort (BB-01-04)
1553	PC can't communicate with LoadPort (BB-01-05)
1554	PC can't communicate with LoadPort (BB-01-06)
1555	PC can't communicate with LoadPort (BB-02-01)
1556	PC can't communicate with LoadPort (BB-02-02)
1557	PC can't communicate with LoadPort (BB-02-03)
1558	PC can't communicate with LoadPort (BB-02-04)
1559	PC can't communicate with LoadPort (BB-02-05)
1560	PC can't communicate with LoadPort (BB-02-06)
1561	PC can't communicate with LoadPort (BB-03-01)
1562	PC can't communicate with LoadPort (BB-03-02)
1563	PC can't communicate with LoadPort (BB-03-03)
1564	PC can't communicate with LoadPort (BB-03-04)
1565	PC can't communicate with LoadPort (BB-03-05)
1566	PC can't communicate with LoadPort (BB-03-06)
1567	PC can't communicate with LoadPort (BB-04-01)
1568	PC can't communicate with LoadPort (BB-04-02)
1569	PC can't communicate with LoadPort (BB-04-03)
1570	PC can't communicate with LoadPort (BB-04-04)
1571	PC can't communicate with LoadPort (BB-04-05)
1572	PC can't communicate with LoadPort (BB-04-06)
1573	PC can't communicate with LoadPort (BB-05-01)
1574	PC can't communicate with LoadPort (BB-05-02)
1575	PC can't communicate with LoadPort (BB-05-03)
1576	PC can't communicate with LoadPort (BB-05-04)
1577	PC can't communicate with LoadPort (BB-05-05)
1578	PC can't communicate with LoadPort (BB-05-06)
1579	PC can't communicate with LoadPort (BB-06-01)

1580	PC can't communicate with LoadPort (BB-06-02)
1581	PC can't communicate with LoadPort (BB-06-03)
1582	PC can't communicate with LoadPort (BB-06-04)
1583	PC can't communicate with LoadPort (BB-06-05)
1584	PC can't communicate with LoadPort (BB-06-06)
1585	PC can't communicate with LoadPort (BB-07-01)
1586	PC can't communicate with LoadPort (BB-07-02)
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1590	PC can't communicate with LoadPort (BB-07-06)
1591	PC can't communicate with LoadPort (BB-08-01)
1592	PC can't communicate with LoadPort (BB-08-02)
1593	PC can't communicate with LoadPort (BB-08-03)
1594	PC can't communicate with LoadPort (BB-08-04)
1595	PC can't communicate with LoadPort (BB-08-05)
1596	PC can't communicate with LoadPort (BB-08-06)